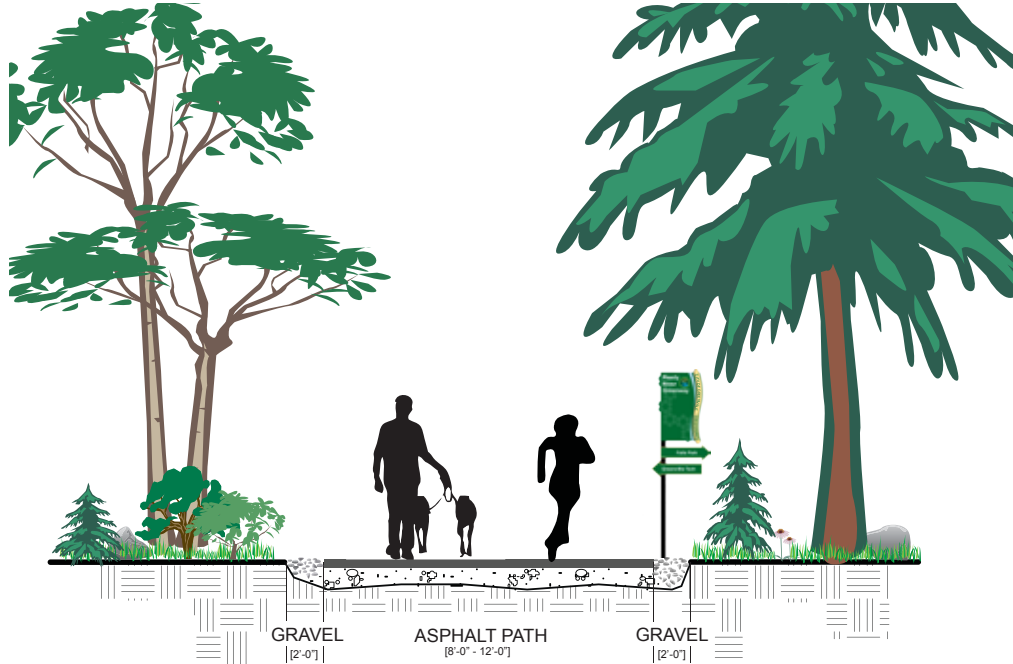


Paved Multi-use Trail: Overview

Multi-use paths are completely separated from motorized vehicular traffic and are constructed in their own corridor, often within an open-space area. Multi-use trails typically have a paved asphalt surface and are capable of being constructed within flood-prone landscapes as well as upland corridors.

- Paved asphalt, concrete or permeable paving is recommended.
 1. In most areas, paved asphalt trails offer substantial durability for the cost of installation and maintenance. As a flexible pavement, asphalt should also be considered for installing a paved trail on slopes.
 2. In areas prone to frequent flooding, and for intensive urban applications, it is recommended that concrete be used for its superior durability and lower maintenance requirements; Consider using high albedo pavement in place of conventional concrete surfaces (it reflects sunlight, reducing radiated heat).
 3. Consider the following for permeable paving: a) It can be twice the cost of asphalt, b) A maintenance schedule for vacuuming debris is required to retain permeability, and c) Not suitable in the floodplain, or in areas without proper drainage (sheet flow or pooling of water with sediment clogs pours).
- Proper trail foundation will increase the longevity of the trail; two inches surfacing material over four inches (min.) of base course gravel over geotextile fabric is recommended. Soil borings may need to be conducted to determine adequate material depths; it should be designed to withstand the loading requirements of occasional maintenance and emergency vehicles.
- Typically 10’ wide, 2% cross slope, with two-foot wide graded shoulders; the shoulders help prevent edges from crumbling and provide an alternate walking and jogging surface.
- Centerline stripes should be considered for trails that generate substantial amounts of traffic, and are particularly useful along curving sections of trail.
- Trail landscaping and maintenance should enhance conditions for wildlife by planting only native species in the trail corridor, removing invasive species when possible, and avoiding harmful pesticides and herbicides. The overall shape of protected natural landscapes along trail corridors also influences wildlife: single, large, contiguous natural areas are more beneficial to wildlife than the same acreage split into smaller segments.



Paved Multi-use Trail: Urban Waterways

‘Paved Multi-use Trail’ guidelines apply, with the following considerations and exceptions:

- Located only in urban areas, where right-of-way constraints and channelized streams restrict trail development to the floodway.
- Typically positioned directly adjacent to the stream channel and are therefore subject to frequent flooding; require hard paved surfaces of concrete to withstand high-velocity stream flows.
- Parking areas near urban waterways can also be retrofitted to accommodate this type of trail.
- When box culverts are built along creeks on planned trail routes, they should be designed to meet with this trail type, and should have sufficient space for trail users.
- Retaining walls or other structural elements may also be required for stable construction and to protect the trail from erosion and flood damage.
- The installation of railings, benches, signage, and trash receptacles, that could obstruct flow during storm events, should be carefully considered.
- The use of retaining walls as seat walls is one way in which non-obtrusive amenities can be included.
- Special consideration should be paid to the mitigation of impacts from trail construction on the natural environment.

Design Guidelines



Paved Multi-use Trail : Floodway Areas

‘Paved Multi-use Trail’ guidelines apply, with the following considerations and exceptions:

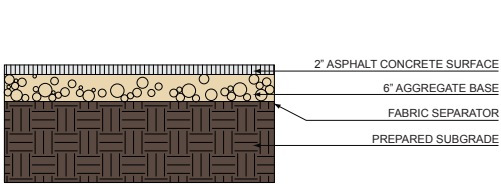
- Typically positioned within the floodway, but not directly adjacent to streams; some vegetative buffer between the stream and trail should be left intact.
- Use existing cleared corridors for trail routing whenever possible, to avoid unnecessary vegetative clearing.
- Subject to infrequent, periodic flooding.
- Require paved surfaces of either asphalt or concrete depending on frequency of flooding and expected velocity of flow.
- No soft shoulder should be constructed due to flood considerations.

- All elements of the trail, including the trail tread, railings, benches, and trash receptacles, will be periodically flooded; design and materials should be carefully selected and sited accordingly.
- Special consideration should be paid to the mitigation of impacts from trail construction on the natural environment.

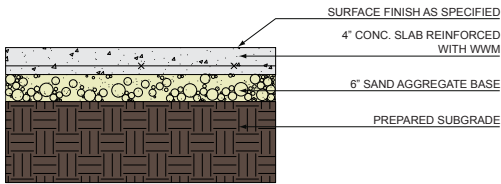


LOW VELOCITY

HIGH VELOCITY



ASPHALT PAVING ON AGGREGATE BASE



CONCRETE PAVING ON AGGREGATE

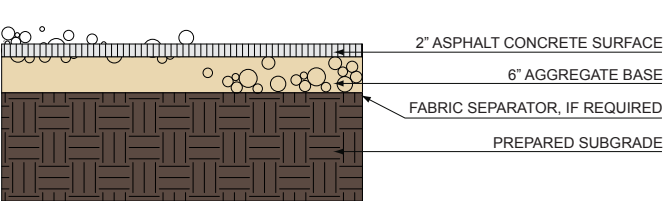
Multi-use Trail : Floodplain Areas

‘Paved Multi-use Trail’ guidelines apply, with the following considerations and exceptions:

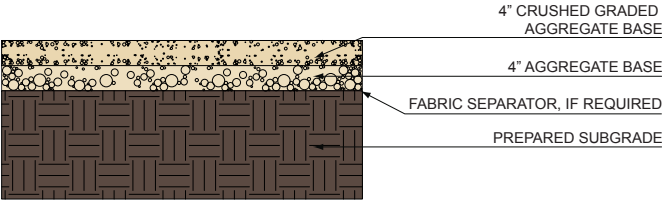
- Typically positioned outside the floodway, within the floodplain; significant vegetative buffer between the stream and trail should be left intact.
- Use existing cleared corridors for trail routing whenever possible, to avoid unnecessary vegetative clearing.
- Subject to occasional flooding, during large storm events.
- Paved asphalt recommended, though an aggregate stone surface may be adequate in some locations.



TYPICAL PAVED & UNPAVED TRAIL CROSS SECTIONS



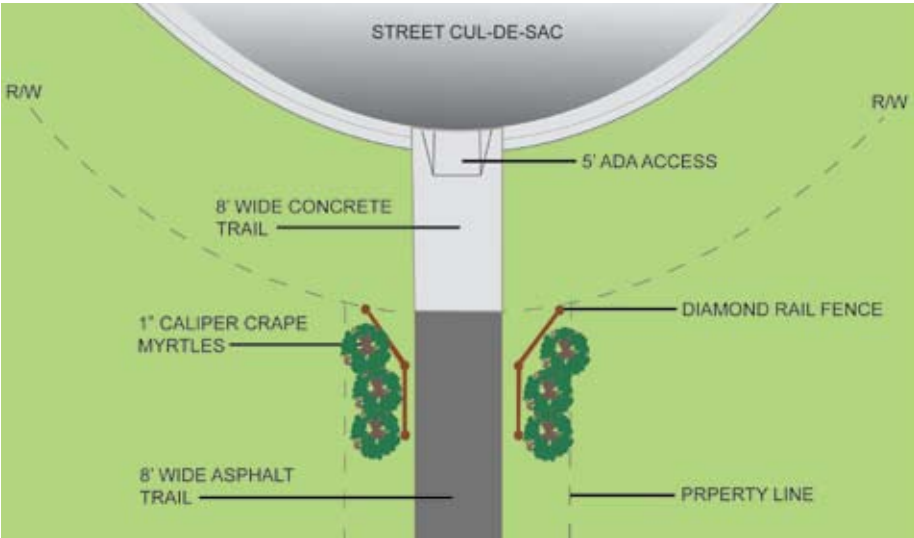
ASPHALT PAVING ON AGGREGATE BASE



GRAVEL PAVING ON AGGREGATE

Design Guidelines

Neighborhood Spur Trail



Neighborhood entrance trail diagram.

Neighborhood spur trails provide residential areas with direct bicycle and pedestrian access to parks, trails, greenspaces, and other recreational areas. They most often serve as small trail connections to and from the larger trail network, typically having their own rights-of-way and easements. Additionally, these smaller trails can be used to provide bicycle and pedestrian connections between dead-end streets, culs-de-sac, and access to nearby destinations not provided by the overall street network. Neighborhood and homeowner association groups are encouraged to identify locations where such connects would be desirable.

- Neighborhood spur trails should remain open to the public.
- Trail pavement shall be at least 8’ wide to accommodate emergency and maintenance vehicles, meet ADA requirements and be considered suitable for multi-use.
- Trail widths should be designed to be less than 8’ wide only when necessary to protect large mature native trees over 18” in caliper, wetlands or other ecologically sensitive areas.
- Access trails should meander whenever possible.
- Landscaping shall be included at the street frontage of the access trail based upon input from the residents of the cul-de-sac or dead-end street. If the access is not in a cul-de-sac, the adjacent property owners and property owners directly across from the access trail will be invited to provide landscape design input. See following section related to landscaping.
- Two sections of diamond rail fencing should be included on each side of the trail near the street frontage. Diamond rail will not be included if the respective neighborhood deeds and covenants do not permit it.

Example of a neighborhood entrance trail, featuring landscape signage.



Vegetation Buffer, Landscaping, and Street Trees

Design Guidelines

Vegetated buffers are used to separate trails not only for floodplain protection and noise from the road, but also, where desired, to screen trail corridors from nearby properties.

- Use native plant species and plants appropriate to the region that are already adapted to the local soil and climate, reducing overall maintenance costs and enhancing local identity. Landscape materials should be installed during the appropriate planting season for the particular species.
- Design the buffer with a combination of evergreen and deciduous plants for year-round interest.
- Plant buffers with a combination of trees and large shrubs, understory plantings, and ground cover.
- Keep the vegetation buffer maintained so that it does not impede views or interfere with trail circulation.
- Avoid vegetation “walls” that box-in trail users.
- Select and place trail vegetation to provide seasonal comfort: shade on trails in the warmer months and warming sunlight on trails in colder months.



Street trees and other plantings provide comfort, a sense of place, and a more natural and inviting setting for pedestrians.

- Street and sidewalk landscaping can be used to provide a separation buffer between pedestrians and motorists, reduce the width of a roadway, calm traffic by creating a visual narrowing of the roadway, enhance the street environment, and help to generate a desired aesthetic.
- Growth pattern and space for maturation, particularly with larger tree plantings, are important to avoid cracking sidewalks and other pedestrian obstructions.
- Islands of vegetation can be created to collect and filter stormwater from nearby streets and buildings. These islands are referred to as constructed wetlands, rain gardens, and/or bioswales. When these devices are employed, the benefits listed above are coupled with economic and ecologic benefits of treating stormwater at its source. See Seattle’s Green Streets Program as a model.

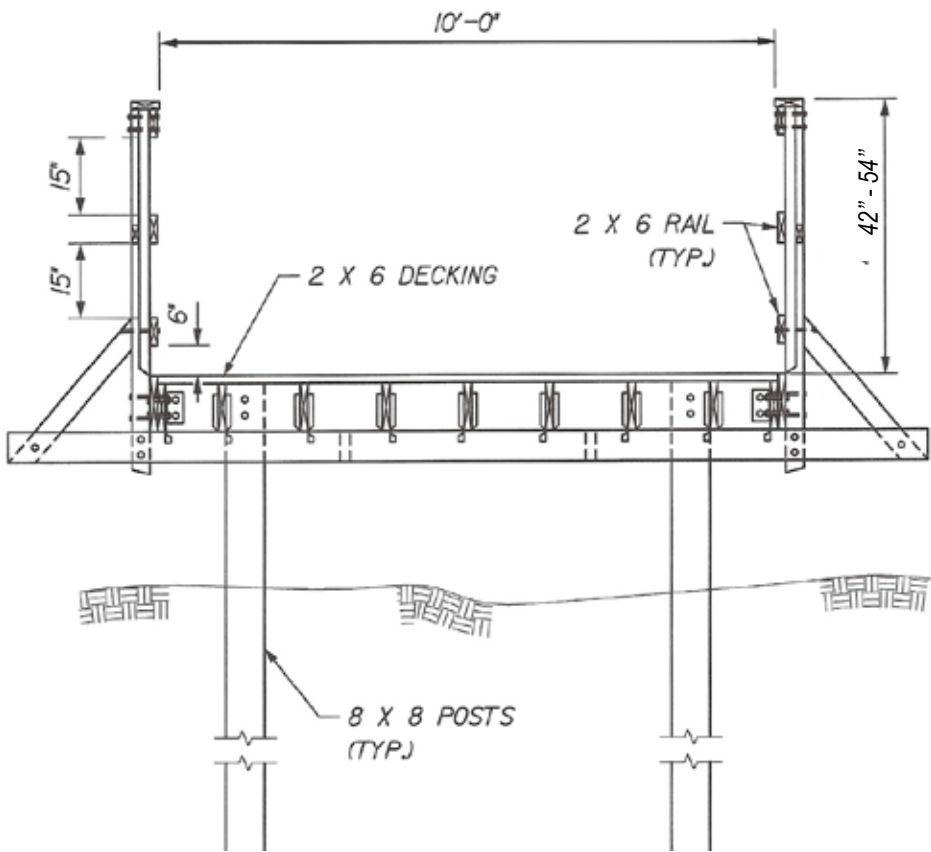


Landscaping used on the Capital Crescent Trail, Washington DC, shows how stormwater treatment can be tied to aesthetically pleasing plantings.

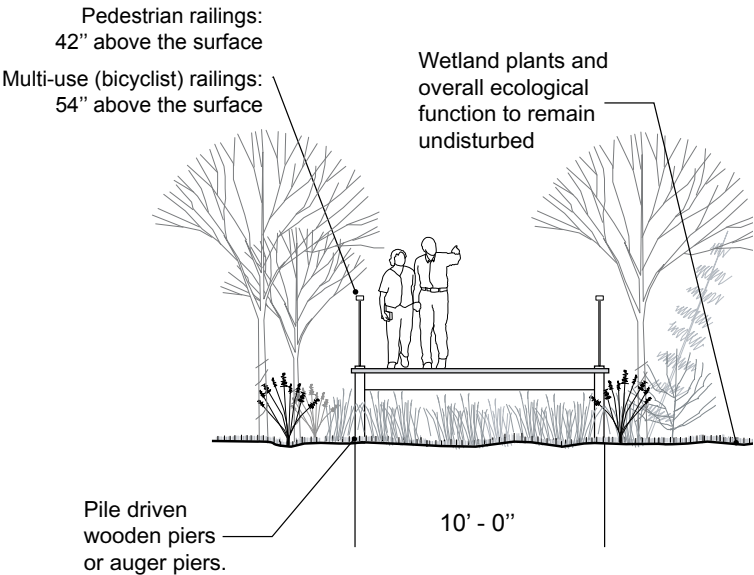
Boardwalk

Boardwalk or wood surface trails are typically required when crossing wetlands or other poorly drained areas. They are constructed of wooden planks or recycled material planks that form the top layer of the boardwalk. The recycled material has gained popularity in recent years since it lasts much longer than wood, especially in wet conditions. A number of low-impact support systems are also available that reduce the disturbance within wetland areas to the greatest extent possible.

- When the height of a boardwalk exceeds 30", railings are required (see section on 'Railings and Fences' for details)
- The thickness of the decking should be a minimum of 2"
- Decking should be either non-toxic treated wood or recycled plastic.
- The foundation normally consists of wooden posts or auger piers (screw anchors). Screw anchors provide greater support and last much longer.
- Opportunities exist to build seating and signage into boardwalks.
- In general, building in wetlands should be avoided.
- Note: muddy bicycle tires may be slick on wood surfaces.



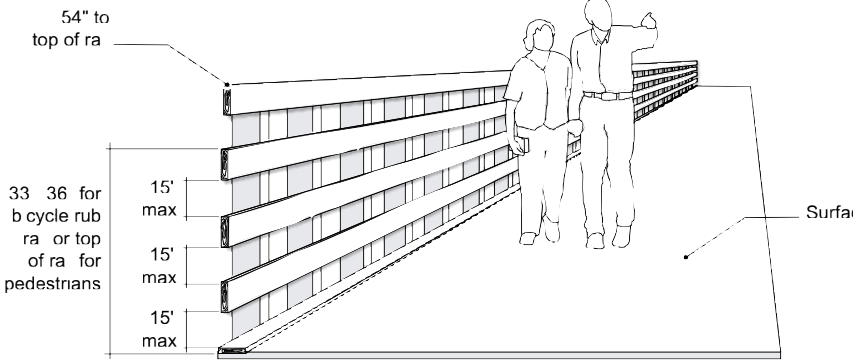
A boardwalk allows for travel through wet areas..



Railings and Fences

Railing and fences are important features on bridges, some boardwalks, or in areas where there may be a hazardous drop-off or hazardous adjacent land uses (such as active rail lines).

- At a minimum, railings and fences should consist of a vertical top, bottom, and middle rail. Picket style fencing should be avoided as it presents a safety hazard for bicyclists.
- A pedestrian railing should be 42-inches above the surface.
- A bicyclist railing should be 54-inches above the surface.
- The middle railing functions as a "rub rail" for bicyclists and should be located 33-and 36-inches above the surface.
- Local, state, and/or federal regulations and building codes should be consulted to determine when it is appropriate to install a railing.



Example image of fence used along a rail with trail (Grand Rounds Parkway).

Innovative Accessways

There are also other innovative ways to provide direct access, particularly in topographically constrained areas (e.g., on steep hills, over waterways, etc.) Stairs, alleyways, bridges, and elevators can provide quick and direct connections throughout the city and can be designed so they are safe, inviting, and accessible to most trail users. For example, stairways can have wheel gutters so that bicyclists can easily roll their bicycles up and down the incline and boardwalks can provide access through sensitive wet areas and across small waterways.



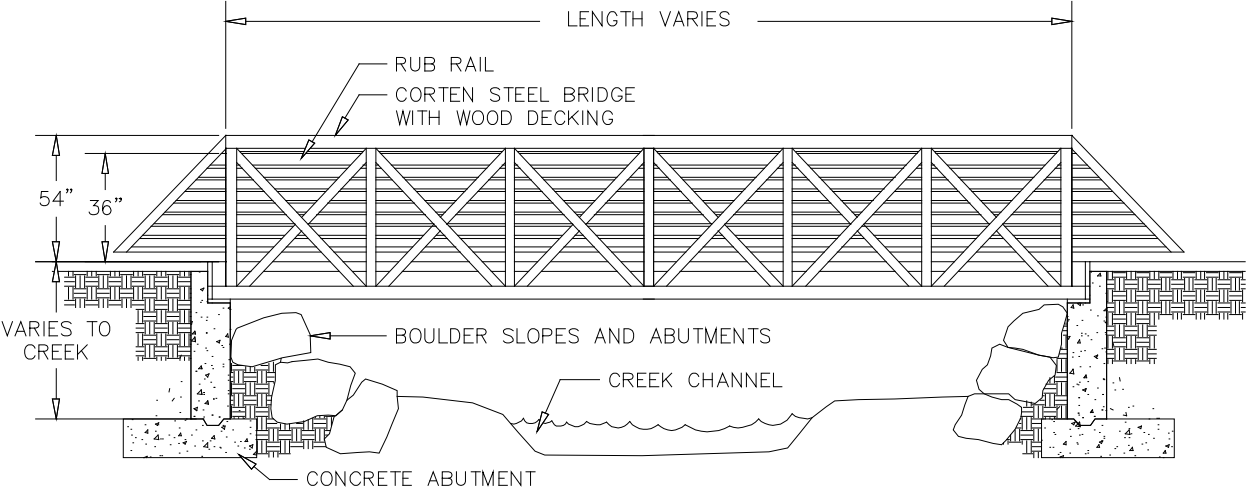
Bicycle wheel gutters on stairs and boardwalk bridge



Trail Bridge

Multi-Use Trail bridges (also ‘bicycle/pedestrian bridges’ or ‘footbridges’) are most often used to provide trail access over natural features such as streams and rivers, where a culvert is not an option. The type and size of bridges can vary widely depending on the trail type and specific site requirements. Some bridges often used for multi-use trails include suspension bridges, prefabricated span bridges and simple log bridges. When determining a bridge design for multi-use trails, it is important to consider emergency and maintenance vehicle access.

- If a corridor already contains a bridge such as an abandoned rail bridge, an engineer should be consulted to assess the structural integrity before deciding to remove or reuse it.
- A trail bridge should support 6.25 tons; Information about the load-bearing capacity of bridges can be found in the American Association of State Highways and Transportation Officials (AASHTO) Standard Specifications for Highway Bridges.
- There are many options in terms of high quality, prefabricated pedestrian bridges available. Prefabricated bridges are recommended because of their relative low cost, minimal disturbance to the project site, and usually, simple installation.
- All abutment design should be sealed by a qualified structural engineer and all relevant permits should be filed.



Trail Overpass

Design Guidelines

Trail overpasses are most often used to provide trail access over large man-made features such as highways and railroads.

- Overpasses work best when existing topography allows for smooth transitions.
- Safety should be the primary consideration in bridge/overpass design.
- Specific design and construction specifications will vary for each bridge and can be determined only after all site-specific criteria are known.
- Always consult a structural engineer before completing bridge design plans, before making alterations or additions to an existing bridge, and prior to installing a new bridge.
- A ‘signature’ bridge should be considered in areas of high visibility, such as over major roadways. While often more expensive, a more artistic overpass will draw more attention to the trail system in general, and could serve as a regional landmark.
- For shared-use facilities, a minimum width of 14’ is recommended.
- Trail overpasses are prohibitively expensive and should only be placed in areas of substantial need.



“Vehicular” Bridges And Underpasses

All new or replacement bridges and tunnels should accommodate pedestrians and bicyclists (except on controlled access roadways where such uses are already prohibited by law). Even though bridge replacements do not occur regularly, it is important to consider these in longer-term pedestrian planning.

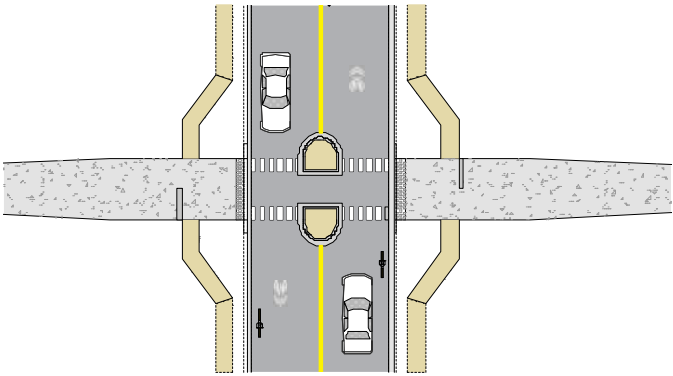
- Sidewalks should be included on roadway bridges on both sides, minimum 5’ wide, with minimum hand-rail height of 42"
- Sufficient bridge deck width should be provided on new bridges, including approaches, to accommodate bicyclists
- In roadway underpasses, where vertical clearance allows, the pedestrian walkway should be separated from the roadway by more than a standard curb height.
- On bridges built for controlled access roadways, a separated, multi-use sidepath should be provided, minimum 12 ‘ wide, with connections made to bike/ped facilities on both sides of the bridge.

Trail Underpass

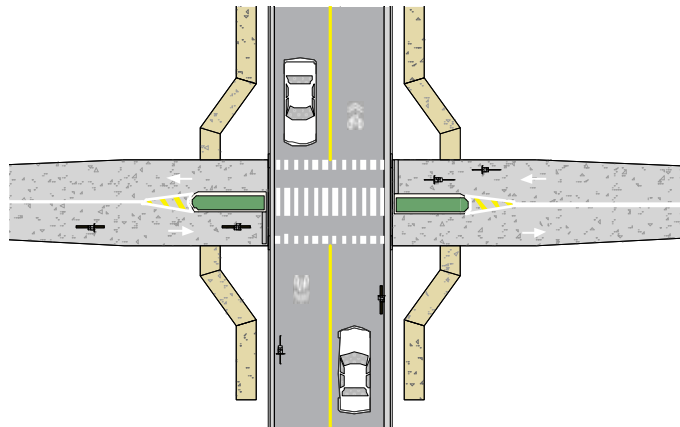
- Over and underpasses should be considered only for crossing arterials with greater than 20,000 vehicle trips per day and speeds 35 - 40 mph and over.
- Underpasses work best with favorable topography when they are open and accessible, and exhibit a sense of safety.
- Underpasses should have a daytime illuminance minimum of 10 fc achievable through artificial and/or natural light provided through an open gap to sky between the two sets of highway lanes, and a night time level of 4 foot-candle.
- Typically utilize existing overhead roadway bridges adjacent to steams or culverts under the roadway that are large enough to accommodate trail users
- Vertical clearance of the underpass is ideally at least 10'; minimum clearance is 8'.
- Width of the underpass is ideally at least 12'; minimum width is 10'.
- Proper drainage must be established to avoid pooling of stormwater, however, some underpasses can be designed to flood periodically (after significant rainfall, for instance).



Trail-Roadway Intersections (Continued)



Median Refuge
Shared Use Path with Sidewalks



Mid-block Crossing
Shared Use Path with Sidewalks and Medians

Trail-Roadway Intersections (Signalized)



- Signalized crossings may be necessary on trails with significant usage when intersecting with demanding roadways, but MUTCD warrants must be met for the installation of a signalized crossing. Consult the MUTCD or NCDOT Division of Bicycle and Pedestrian Transportation for signal, sign and light placement.
- FHWA issued an interim approval for the optional use of rectangular rapid flashing beacons (RRFBs, shown at left) as warning beacons supplementing pedestrian crossing or school crossing warning signs at crossings across uncontrolled approaches. An analysis by the Center for Education and Research in Safety found them to have much higher levels of effectiveness in making drivers yield at crosswalks than the standard over-head and side-mount round flashing beacons.

Trail Heads

Major access points should be established near commercial developments and transportation nodes, making them highly accessible to the surrounding communities. Minor trailheads should be simple pedestrian and bicycle entrances at locally known spots, such as parks and residential developments.

A minor trailhead could include facilities such as parking, drinking fountains, benches, a bicycle rack, trash receptacles, and an information kiosk and/or signage. Major trailheads could include all of the above plus additional facilities, such as rest rooms, shelters, picnic areas, a fitness course, an emergency telephone, and a larger parking area.

Partnerships could also be sought with owners of existing parking lots near trails. Benefits are three fold: Business benefit from trail-user patronage; trail owners benefit from not having to buy more land and construct a parking facility; and the environment benefits from less development in the watershed.



A major trail head at the Capital Crescent Trail in Maryland, featuring concessions and bicycle, canoe, and kayak rentals.



A water fountain and pet-water fountain..



Example layout for a major trail head.



Example layout for a minor trail head.

Design Guidelines

Trail Amenities

Benches

Description: There are a wide variety of benches to choose from in terms of style and materials. The illustrated bench is a custom design that reflects the industrial feel of the warehouse district it is found in. Material selection should be based on the desired design theme as well as cost.

Recommendations: Due to a wide range of users, all benches should have a back rest. A bench should normally be 16 - 20" above ground with sturdy handrails on either side. The seating depth should be 18-20" and the length should vary between 60 - 90". Provide wheelchair access alongside benches, at least a 30-by-48-inch area for adequate maneuvering. If benches are next to each other (either side by side or face to face), allow 4 feet between them.



Other Seating

Description: Other more informal seating opportunities may exist along a trail or near a parking area where other furniture like a picnic table may be appropriate.

Recommendations: This type of furniture can be triangulated with cooking facilities, and a trash receptacle. Wheelchair access spacing recommendations, as noted in the preceding section on ‘benches,’ also applies to other seating.



Trash Receptacles

Description: Trash receptacles should be constructed of a suitable material to withstand the harsh elements of the outdoor environment. Additionally trash receptacles should ensure that litter is contained securely preventing contamination or spillage into the surrounding environment.

Recommendations: Trash receptacles should be placed along the trail and at all trail-heads. Adequate trash receptacles will combat littering and preserve the natural environment for all trail users.



Public Art

Explore opportunities to include public art within the overall design of the trail system. Local artists can be commissioned to provide art for the trail system, making it uniquely distinct. Many trail art installations are functional as well as aesthetic, as they may provide places to sit and play on. According to American Trails,

“Art is one of the best ways to strengthen the connection between people and trails. Across America and elsewhere, artists are employing a remarkably wide range of creative strategies to support all phases of trail activities, from design and development to stewardship and interpretation. In particular, art can be an effective tool for telling a trail’s story compellingly and memorably.”

Example art programs for trails can be found at:
www.americantrails.org/resources/art/ArtfulWays.html

Design Guidelines



Trail Lighting

Lighting for multi-use trails should be considered on a case-by-case basis in areas where 24-hour activity is expected (such as college campuses or downtown areas), with full consideration of the maintenance commitment lighting requires. In general, lighting is not appropriate for off-road trails where there is little to no development.

- A licensed or qualified lighting expert should be consulted before making any lighting design decisions. Doing so can reduce up-front fixed costs as well as long-term energy costs.
- Use full cut-off, energy-efficient lighting that is IDA Approved Dark Sky Friendly to avoid excess light pollution and save costs (See www.darksky.org for more info)
- If a main trail corridor is unlit and closes at dark, extended hours for commuters should be considered, particularly during winter months when trips to and from work are often made before sunrise and after dusk. See the American Tobacco Trail in Durham, NC, as an example, which is unlit and remains open to commuters until 10 PM.
- Consider lighting at the following locations:
 - Entrances and exits of bridges
 - Public gathering areas along the greenway
 - Trail access points
- Only use lighting along a trail if:
 - Night usage is desired or permitted
 - It is acceptable to residents living along or near the trail
 - The area is not a wildlife area

Roadway Lighting

Proper lighting in terms of quality, placement, and sufficiency can greatly enhance a nighttime urban experience as well as create a safe environment for motorists and pedestrians. Two-thirds of all pedestrian fatalities occur during low-light conditions (AASHTO, 2004: Guide for the Planning, Design, and Operation of Pedestrian Facilities). Attention should be paid to crossings so that there is sufficient ambience for motorists to see pedestrians. To be most effective, lighting should be consistent, adequately spaced, and distinguished, providing adequate light.

In commercial or downtown areas and other areas of high pedestrian volumes, lower level, pedestrian-scale lighting with emphasis on crossings and intersections may be employed to generate a desired ambience. Roadway streetlights can range from 20-40 feet in height while pedestrian-scale lighting is typically 10-15 feet. It is important to note that every effort should be made to address and prevent light pollution. Also known as photo pollution, light pollution is ‘excess or obtrusive light created by humans’.

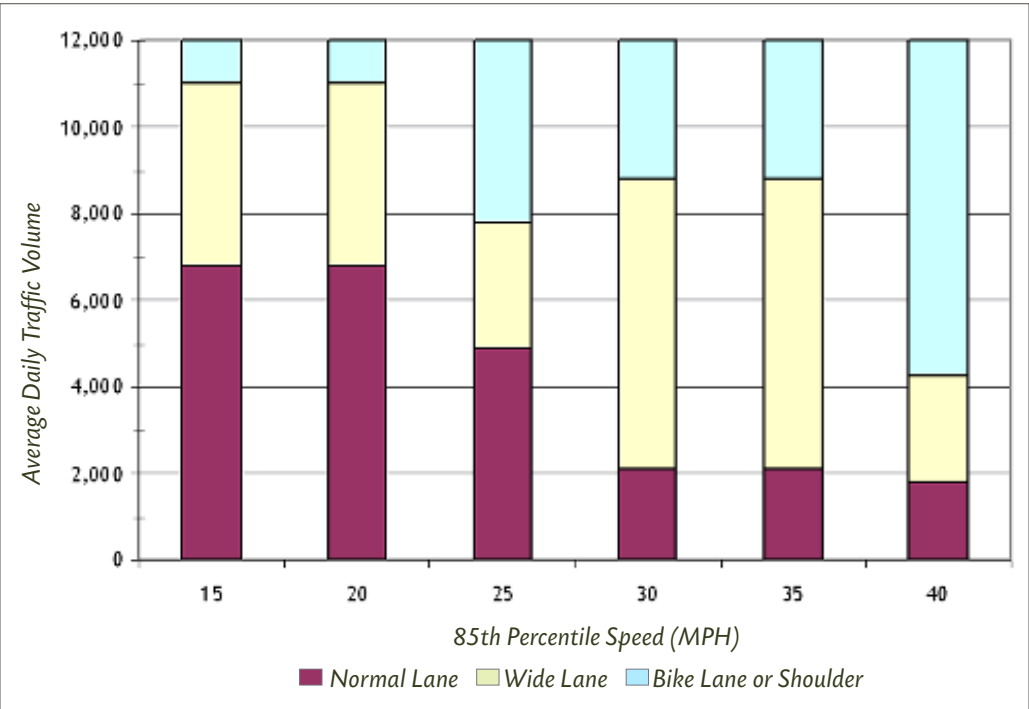
- Ensure pedestrian walkways and crossways are sufficiently lit.
- Consider adding pedestrian-level lighting in areas of higher pedestrian volumes, downtown, and at key intersections.
- Install lighting on both sides of streets in commercial districts.
- Use uniform lighting levels
- Use full cut-off, energy-efficient lighting that is IDA Approved Dark Sky Friendly to avoid excess light pollution and save costs (See www.darksky.org for more info)

On-road Bicycle Facilities

Design Guidelines

A wide variety of on-road bicycle facilities have been developed to meet different transportation needs in different roadway situations. The appropriate bicycle facility for any particular roadway, whether new or existing, should be dictated primarily by vehicle volume and speed of the roadway. The figure below provides a matrix for evaluating bicycle facilities. The speed of the travel lane is shown along the x-axis and total traffic volumes per day are shown along the y-axis. The different colors represent the type of bikeway facility prescribed given the volume and speed of the travel lane.

North American Speed-Volume Chart



Source: M. King: Bicycle Facility Selection: A Comparison of Approaches

Shared Roadways

By state law, bicycles are vehicles and bicyclists have the same rights and responsibilities as motor vehicles drivers, including the right to share travel lanes on all roadways, except limited access highways. There are several types of roadway environments in which bicycles most commonly share the travel lane with other vehicles. Hence, some of these roadway types are also considered bicycle facilities, including neighborhood streets, bicycle boulevards, wide outside lanes, and streets with shared-lane markings.

Neighborhood Streets

Many bicyclists can safely share the road with vehicles on low volume (less than 3,000 cars per day), low speed roadways (e.g., a residential or neighborhood street).



Left: Neighborhood street examples.

Bicycle Boulevards

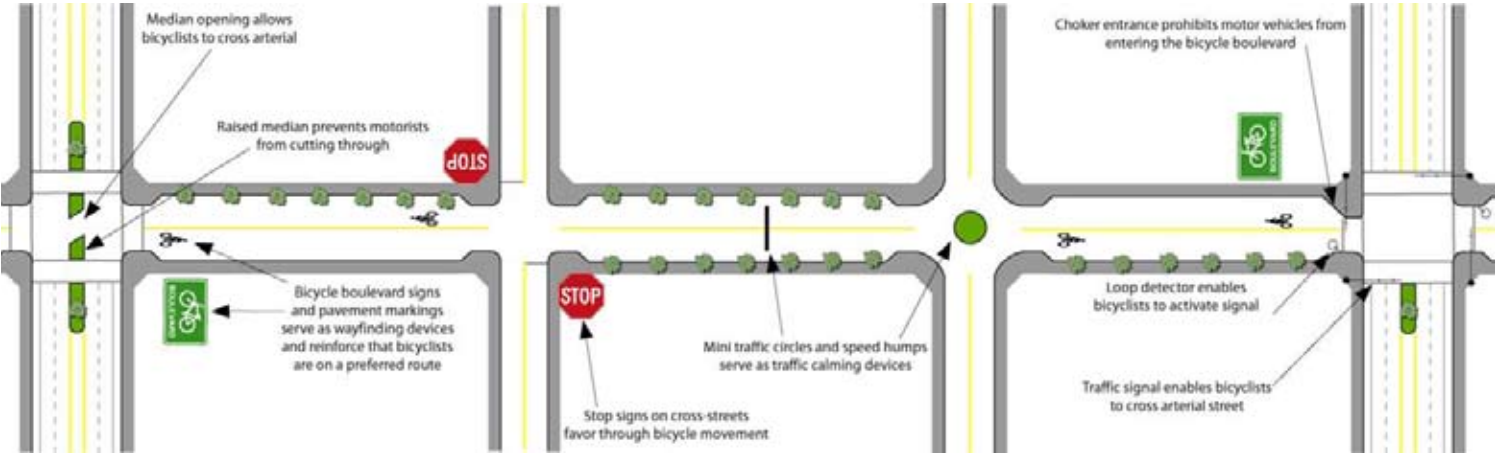
To further identify preferred routes for bicyclists, the operation of lower volume roadways may be modified to function as a through street for bicycles while maintaining local access for automobiles. Traffic calming devices reduce traffic speeds and through trips while limiting conflicts between motorists and bicyclists, as well as give priority to through bicycle movement.

For a complete overview, see www.ibpi.usp.pdx.edu/guidebook.php



Above: Bike boulevard route pavement markings and signs direct bicyclists.

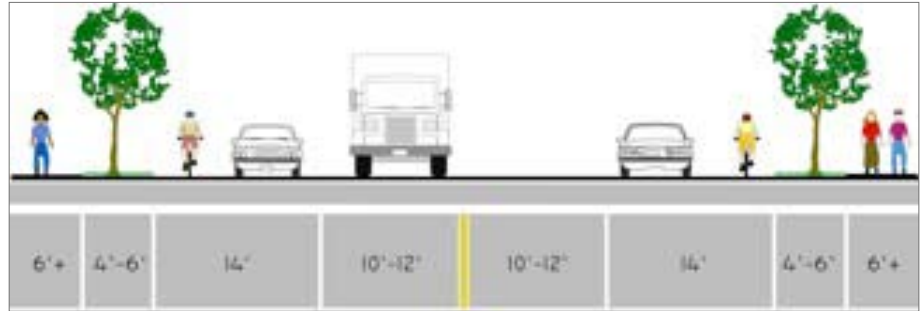
Below: A bicycle boulevard.



Wide Outside Lanes

Even without a bicycle facility or marking, the conditions for bicyling are improved when the outside travel lane in either direction is widened to provide enough roadway space so that bicyclists and motor vehicles can share the roadway without putting either in danger (e.g., higher volume roadways with wide (14') outside lanes).

Below: Wide Outside Lane on a Typical Two Lane Roadway

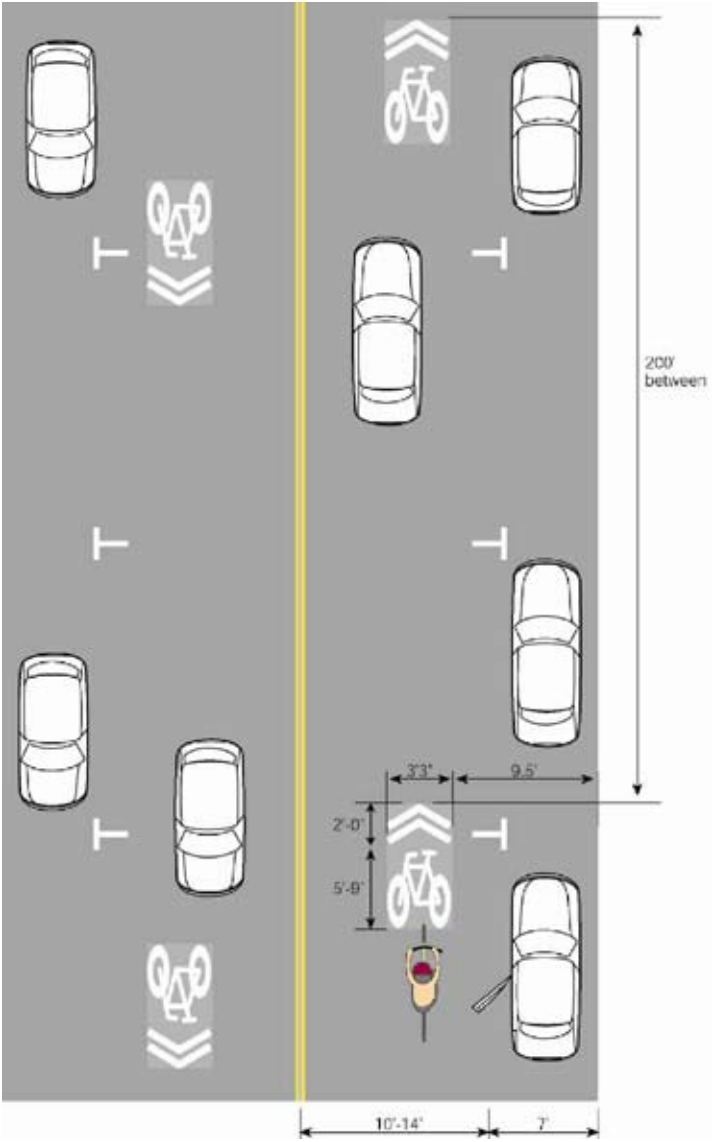


Shared Lane Marking

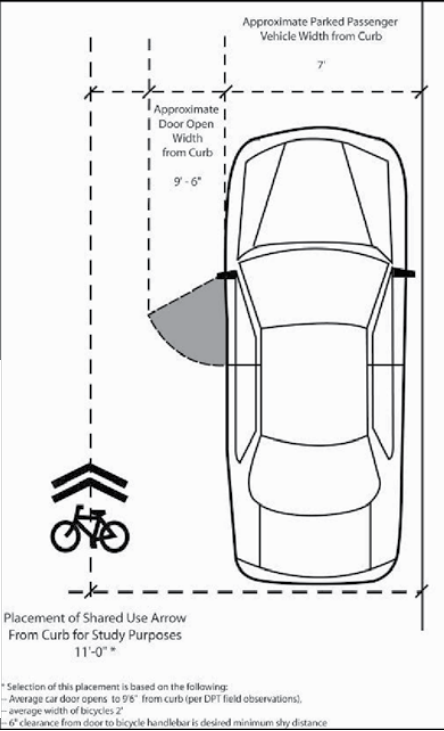
A bicycle shared lane marking (or 'sharrow') can serve a number of purposes, such as making motorists aware of bicycles potentially traveling in their lane, showing bicyclists the appropriate direction of travel, and, with proper placement, reminding bicyclists to bike further from parked cars to prevent "dooring" collisions. The shared lane marking stencil is used:

- Where lanes are too narrow for striping bike lanes
- Where the speed limit does not exceed 35 MPH
- With or without on-street parking

For one-way streets with two travel lanes, the marking should be placed in the left lane to reduce chances of a dooring collision. Even though traffic laws generally call for slower vehicles in the right-hand lane, there is an exception for bicyclists ("Any person operating a pedalcycle upon a roadway which carries traffic in one direction only and has two or more marked traffic lanes may ride as near the left-hand curb or edge of the roadway as practicable" www.dmv.state.pa.us/pdotforms/vehicle_code/chapter35.pdf)



Design Guidelines



Shared lane markings in Bethlehem, PA.

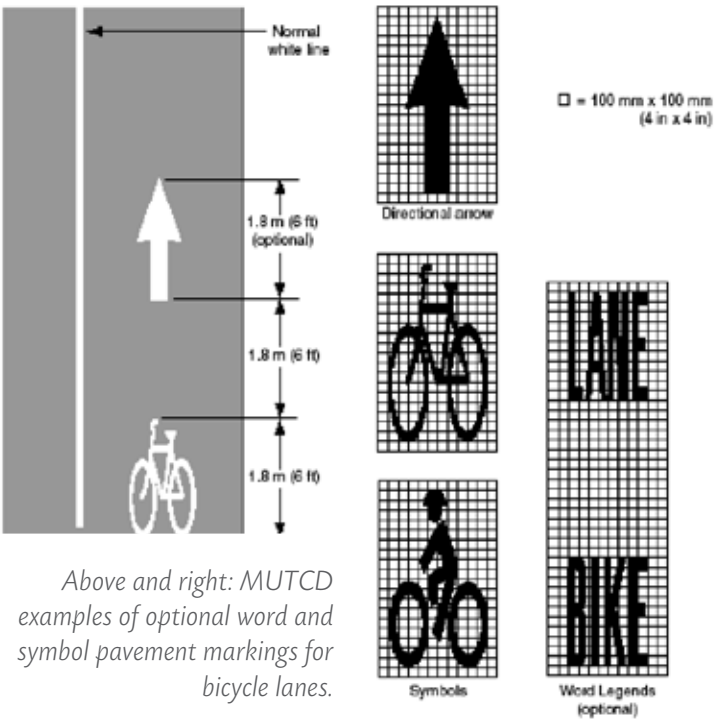


Bicycle Lanes

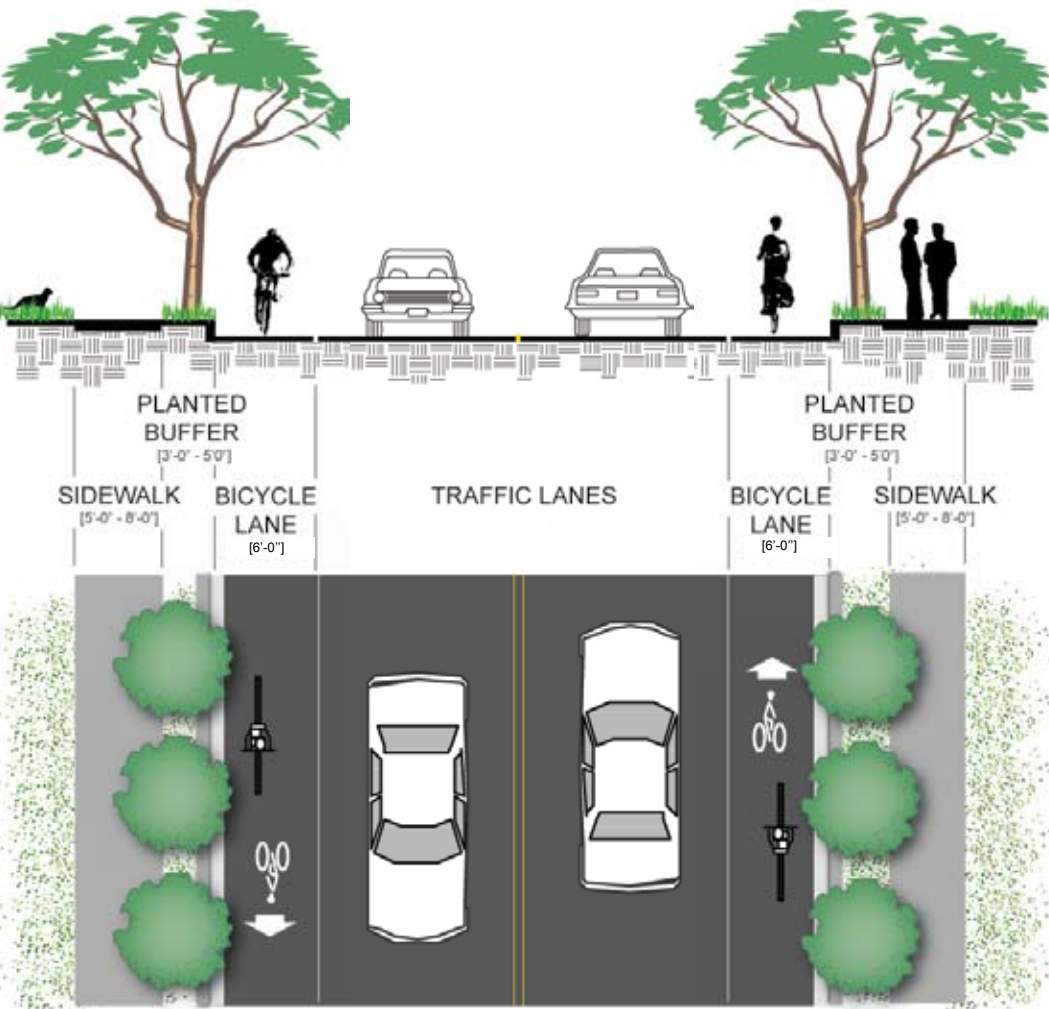
A bicycle lane is a portion of the roadway that has been designated by striping, signing, and pavement markings for the preferential and exclusive use of bicyclists. Bicycle lanes are located on both sides of the road, except one way streets, and carry bicyclists in the same direction as adjacent motor vehicle traffic. In the City of Allentown, PA, many local cyclists have expressed the desire to use striped shoulders as an alternative to bicycle lanes (see guidelines for ‘Striped/Paved Shoulders’).

Recommended bicycle lane width:

- 6’ from the curb face when a gutter pan is present (or 4’ from the edge of the gutter pan)
- 4’ from the curb face when no gutter pan is present
- Should be used on roadways with 3,000 or more ADT
- Not suitable where there are a high number of commercial driveways
- Suitable for 2-lane facilities and 4-lane divided facilities



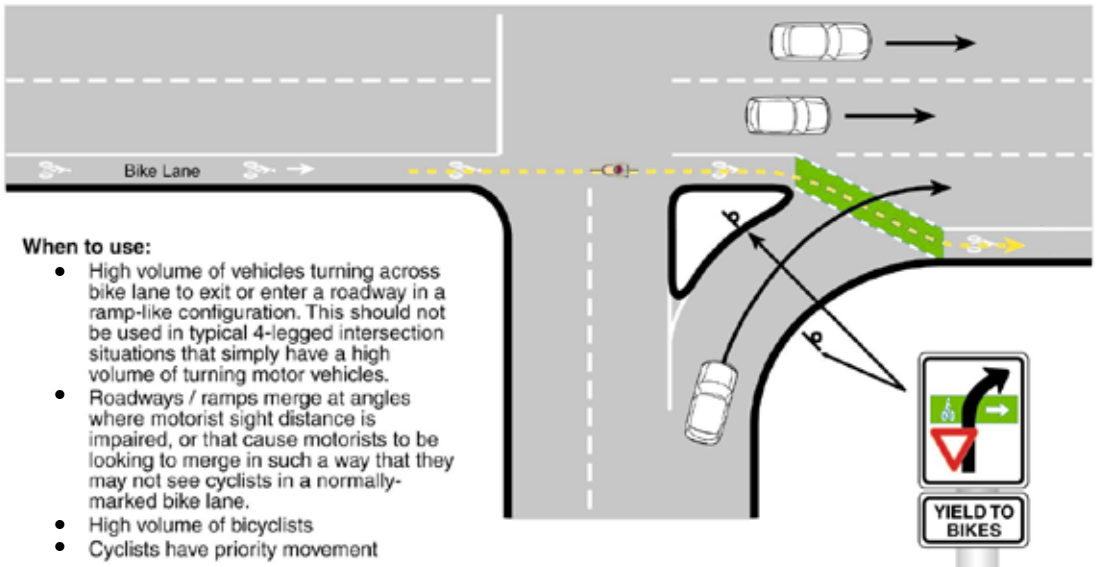
Above and right: MUTCD examples of optional word and symbol pavement markings for bicycle lanes.



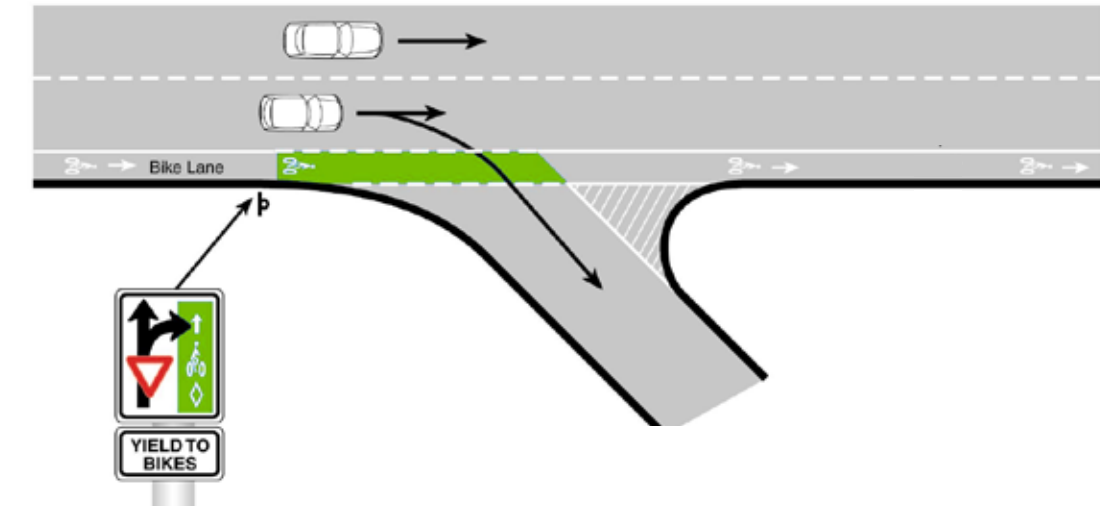
Colored Bike Lanes

In addition to markings presented in the MUTCD, the following experimental pavement markings may be considered. European countries have used colored pavement for bicycle lanes in areas that tend to have a higher likelihood for vehicle conflicts. Examples of such locations are freeway on- and off-ramps and where a motorist may cross a bicycle lane to move into a right turn pocket. In the United States, the City of Portland and New York City have colored bike lanes and supportive signing with favorable results. Studies after implementation showed more motorists slowing or stopping at colored lanes and more motorists using their turn signals near colored lanes. Green is the recommended color (some cities that have used blue are changing to green, since blue is associated with handicapped facilities).

Entrance Ramp Zone



Exit Ramp Zone



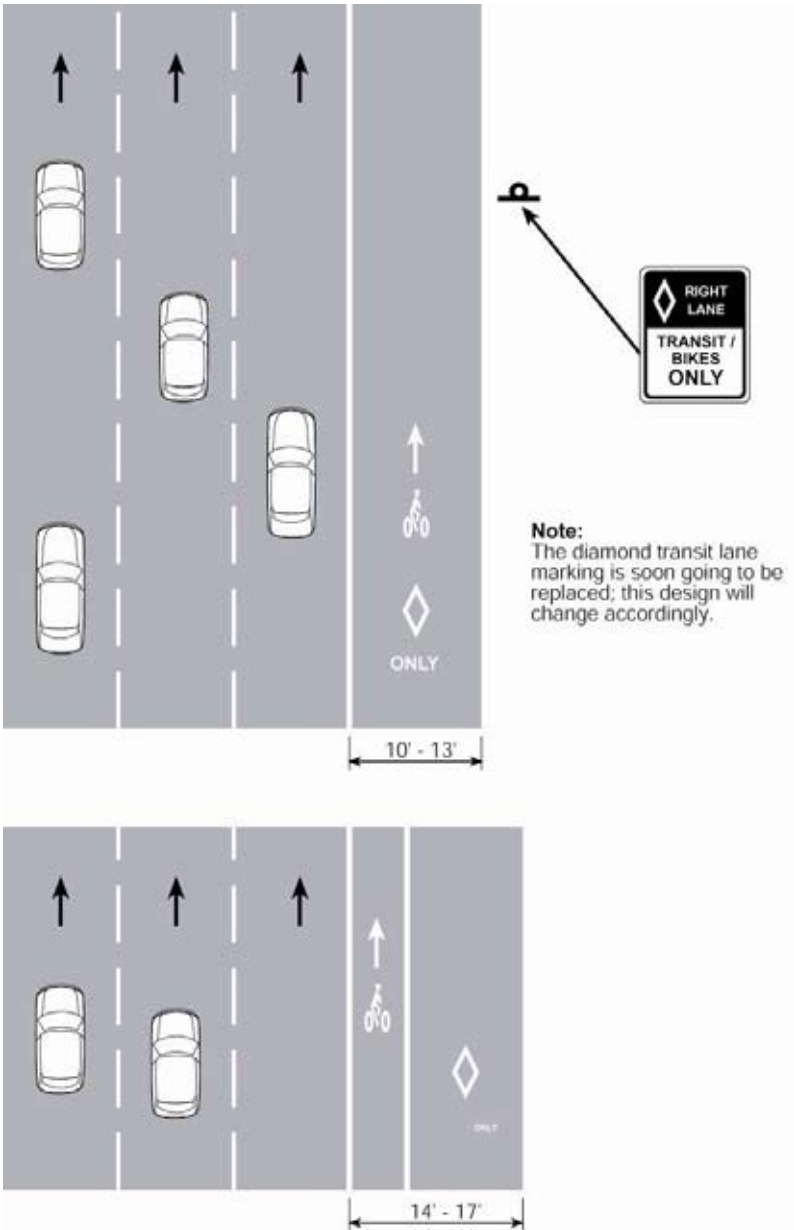
Henry Street in Brooklyn, NY.

Shared Bus/Taxi/Bicycle Lane

Many cities have created multi-use lanes that accommodate bicycles and transit vehicles within the street. This innovative bikeway treatment is utilized in Phoenix, AZ; Philadelphia, PA; and Toronto, Canada.

Potential applications include:

- On auto-congested streets with moderate or long bus headways
- Moderate bus headways during peak hour
- Areas with limited alternative routes

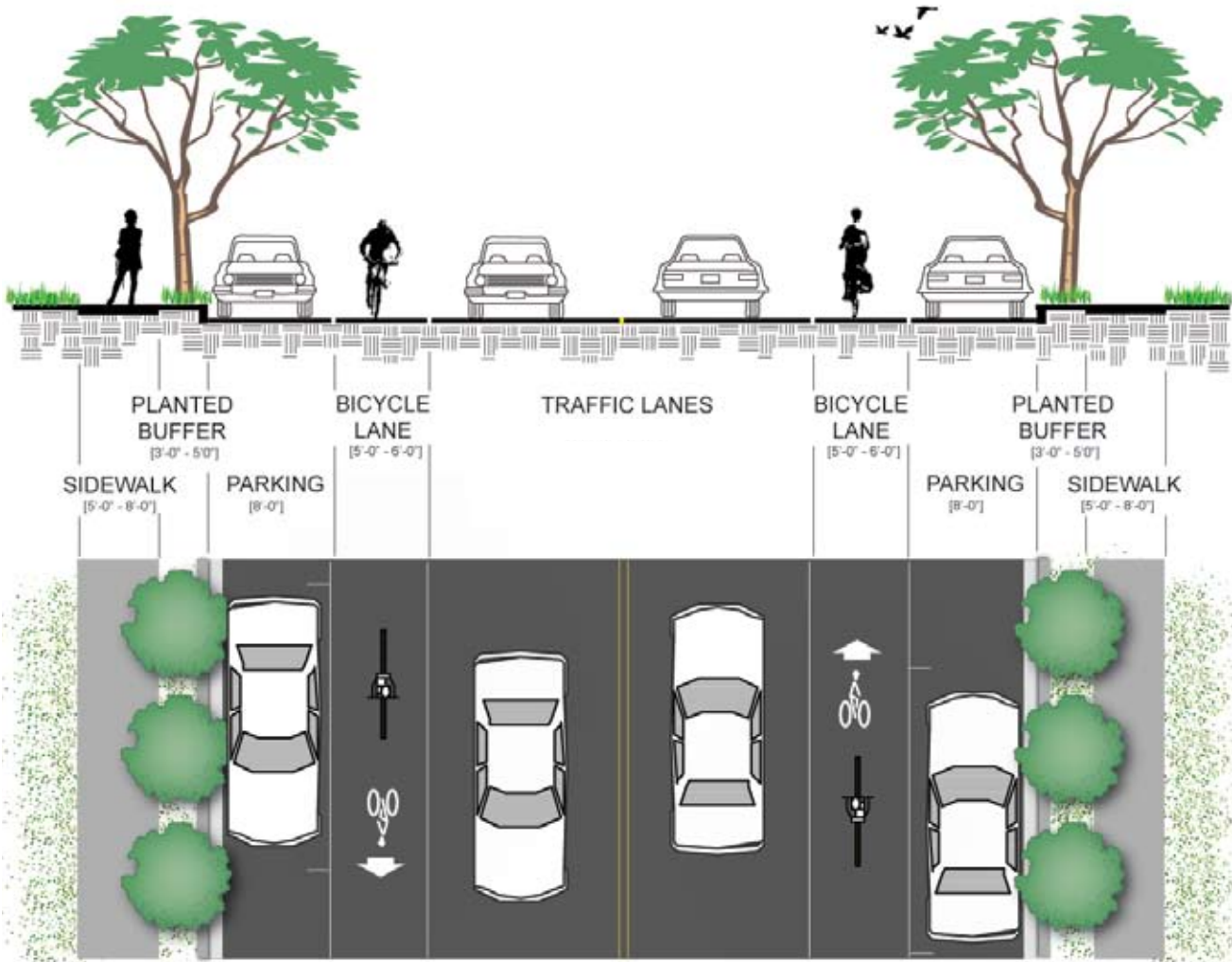


Bicycle Lane + Parking

Design Guidelines

On moderate volume roadways, such as minor collectors, on-street parking is often permitted. Where on-street parking is permitted, and a bike lane is provided, the bike lane must be between parking and the travel lane, and be a minimum of 6' wide.

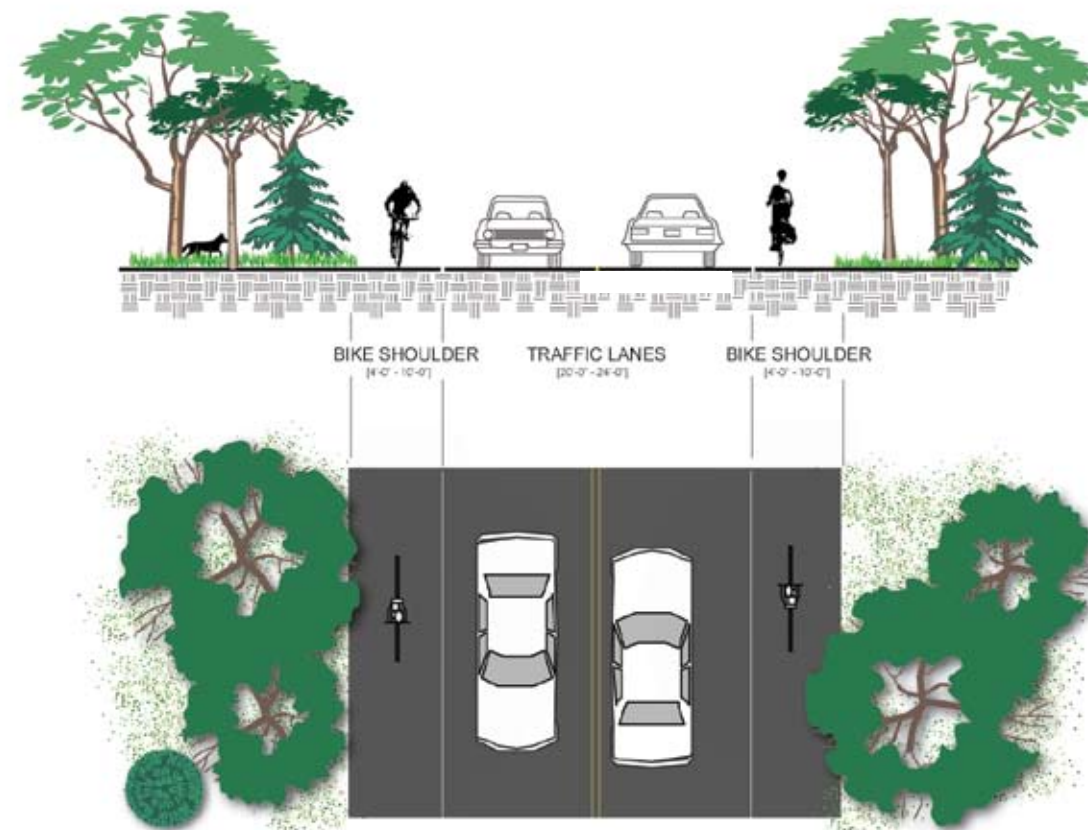
- Appropriate space must be allocated to allow passing cyclists room to avoid open car doors.
- For lanes with combined vehicle parking and bike use (as shown below, in the photo at left), a minimum width of 12' to 13' is recommended, and AASHTO recommends 11' to 13'.



Striped/Paved Shoulder

Paved shoulders are the part of a roadway which is contiguous and on the same level as the regularly traveled portion of the roadway. There is no minimum width for paved shoulders, however a width of at least four feet is preferred. Ideally, paved shoulders should be included in the construction of new roadways and/or the upgrade of existing roadways, especially where there is a need to more safely accommodate bicycles.

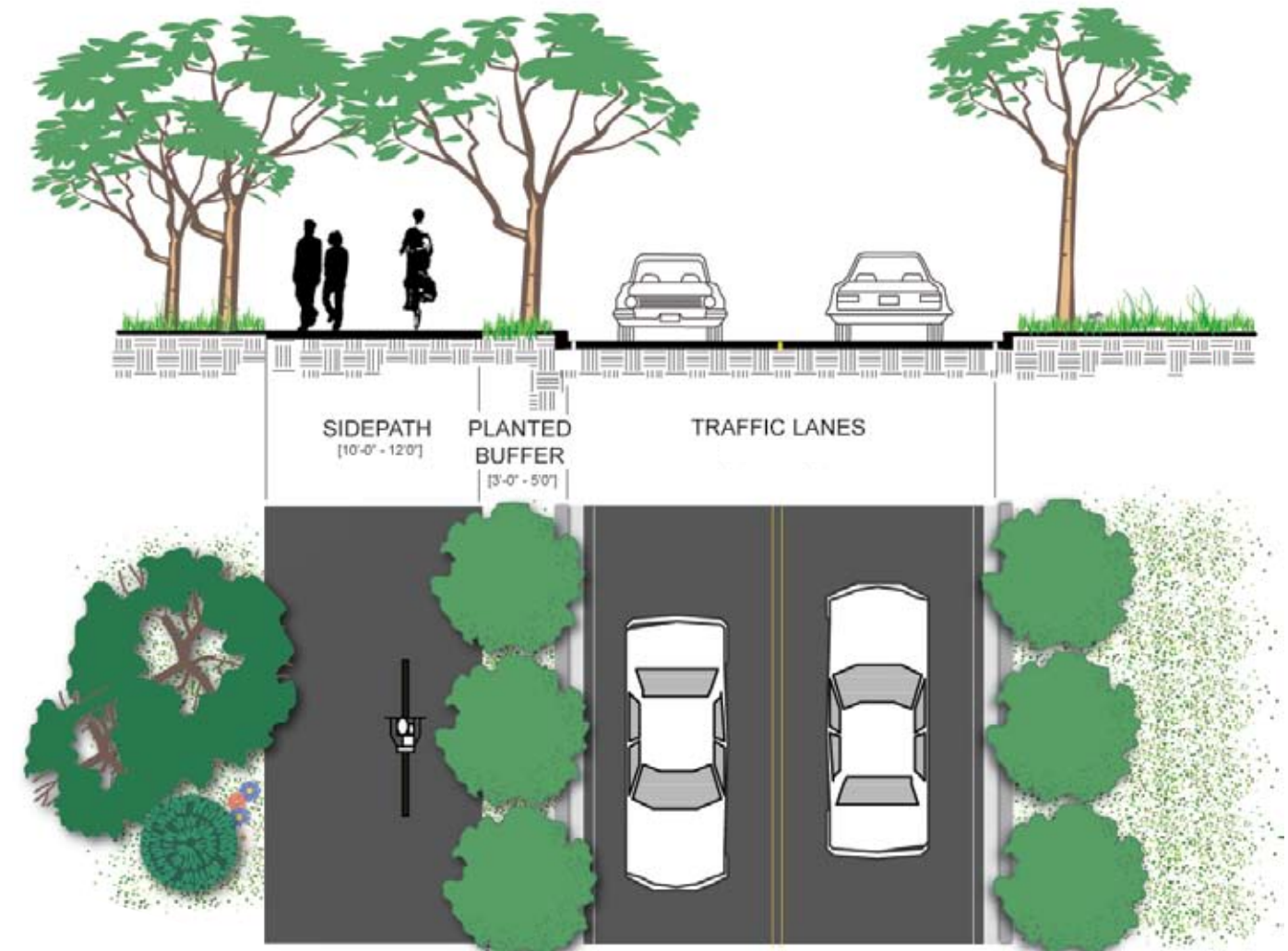
- Most often used in rural environments, although not confined to any particular setting
- Should be delineated by a solid white line, and provided on both sides of the road
- Should be contiguous and on the same level as the regularly traveled portion of the roadway
- 4' minimum width; however for speeds higher than 40 MPH with high ADT, a shoulder width of more than 4' is recommended.
- Rumble strips should be avoided, but if used, then a width of more than 4' is needed.
- Paved shoulders should not be so wide as to be confused with a full automobile travel lane



Sidepath

Multi-use paths located within the roadway corridor right-of-way, or adjacent to roads, are called 'Sidepaths'. Sidepaths provide a comfortable walking space for pedestrians and enables children and recreational bicyclists to ride without the discomfort of riding in a busy street.

- This configuration works best along roadways with limited driveway crossings and with services primarily located on one side of the roadway, or along a riverfront or other natural feature.
- A minimum 10' width is necessary on sidepaths for bicyclists to pass one another safely (12' for areas expecting high use)
- A 6' or greater vegetated buffer between the sidepath and the roadway should be provided where possible.
- Roadway corridors where side paths are recommended should also have adequate on-road bicycle facilities (such as shared lane markings, paved shoulders, or bicycle lanes), so that all levels of bicyclists are accommodated.
- Well-designed transitions from sidepaths to on-road facilities will direct bicyclists to the correct side of the roadway (see guidelines for Trail-Roadway Intersections)



Design Guidelines

Green Alleyways

Green alley projects are being implemented in cities all over the U.S. and Canada in an effort to retrofit alleys to reduce runoff and better absorb rainwater through the use of permeable pavements, landscaping and infiltration basins. The best resource currently available (online) is the City of Chicago's 2009 Green Alley Handbook, which explains why the city is interested in sustainable alley design, illustrates the BMP techniques the City will use in green alley design, and provides sample layouts of how these elements have been combined in pilot applications. In addition, information and resources are provided for property owners interested in implementing their own environmental BMPs, ranging from recycling to installing your own rain gardens. The five main elements for green alleys outlined in the handbook include: 1) Alley Drainage Improvement through Proper Alley Pitching and Grading; 2) Permeable Pavement; 3) High Albedo Pavement; 4) Recycled Construction Materials; and 5) Dark Sky Compliant Light Fixtures.

Many cities are taking these initiatives a step further, aiming to attract pedestrians and bicyclists to these newly renovated, low-volume corridors. Considerations for using green alleyways as bicycle and pedestrian facilities include the following:

- The top consideration for using a green alley as bicycle and pedestrian corridor is whether the retrofitted alleyway would serve as a superior connection compared with nearby streets. The best candidates for alleyways as bicycle facilities are ones that provide a continuous, parallel route to a street that is both dangerous for cyclists, and is unlikely to accommodate bicycles in the future (due to costs or other constraints).
- To increase circulation space for bicycles and pedestrians in green alleys, parking in the alley should only be allowed for alley residents without driveways or garages directly off the alley. For parking under these circumstances, the alleyway should be designed with trees and/or tree planters to help organize the parking, making it a more pleasant environment for cyclists (see woonerf parking example at right).
- The green alley design option of replacing a center strip of pavement with grass is not recommended for alleys intended for bicycle use; a permeable pavement center strip is recommended as an alternative to this option.
- Automobile traffic on green alleys should be limited to alley residents accessing their property.



Design Guidelines

- Sufficient lighting should be made available at night and on dark days. See section on 'Lighting' for related guidelines.
- Where green alleys intersect streets, integrate features into the streets that would prioritize bicyclists (ie improve crossing safety) and improve driver awareness as to presence of bikes. See section on 'Bicycle Boulevards' for more related guidelines.
- Ensure that green alleys are plowed in the winter, providing year round accessibility.
- Generating funding for alley greening projects is most feasible through public-private partnerships and various stormwater-related state and federal programs.
- Surface should be smooth and clear of debris.



The concept of 'green alleyways' is varied and evolving. Many focus primarily on water infiltration (opposite-top), others slow automobile traffic and prioritize bike/ped traffic (opposite-bottom and above), while still others close automobile traffic entirely, creating areas for exclusively for vegetation and people (right).



Bicycle-Friendly Intersections

Intersections represent one of the primary collision points for bicyclists, with many factors involved:

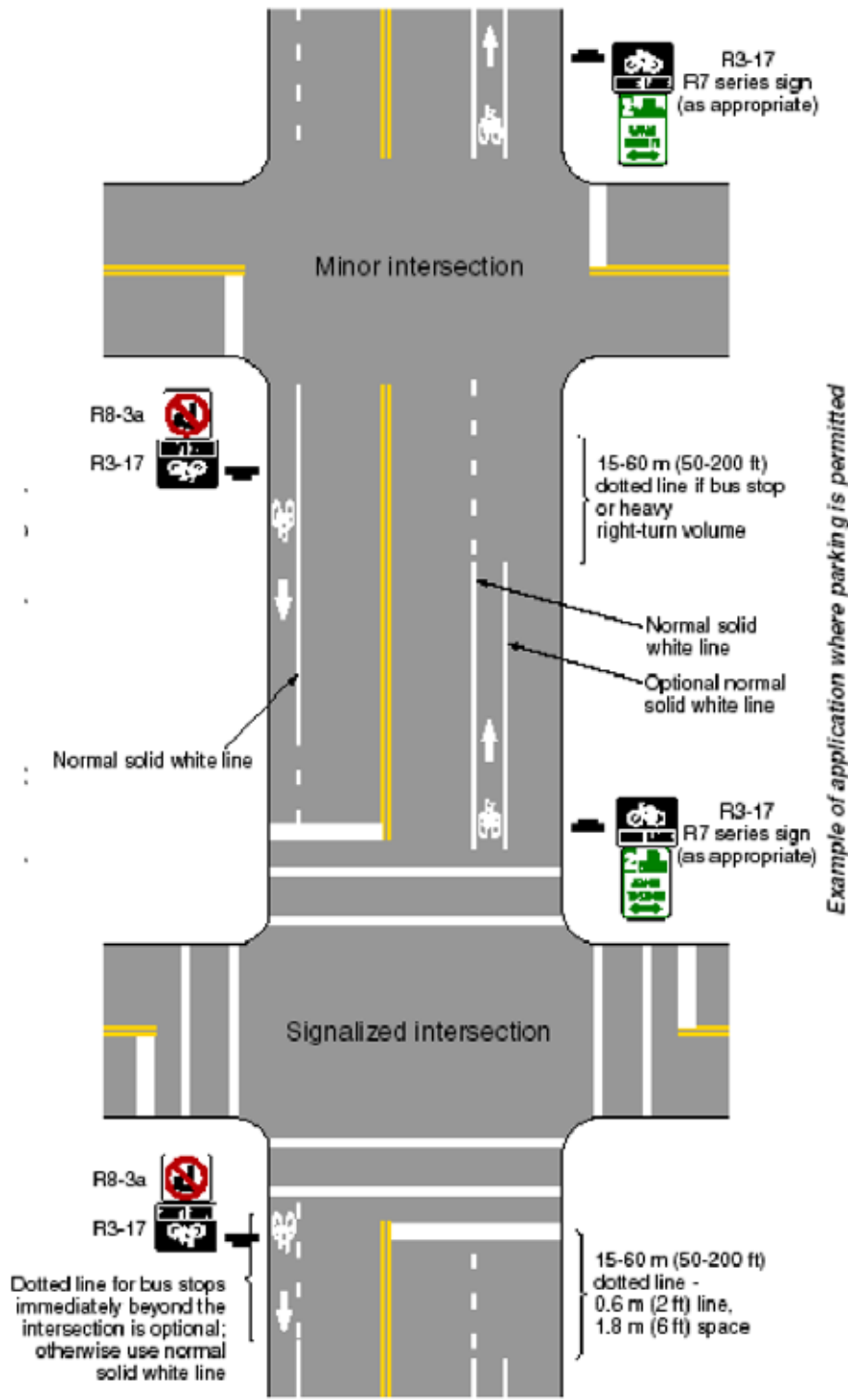
- Larger intersections are more difficult for bicyclists to cross.
- On-coming vehicles from multiple directions and increased turning movements make it more difficult for motorists to notice non-motorized travelers.
- Most intersections do not provide a designated place for bicyclists.
- Bike lanes and pavement markings often end before intersections, causing confusion for bicyclists.
- Loop and other traffic signal detectors, such as video, often do not detect bicycles.
- Bicyclists making a left turn must either cross travel lanes to a left-turn lane, or dismount and cross as a pedestrian.
- Bicyclists traveling straight may have difficulty maneuvering from the far right lane, across a right turn lane, to a through lane of travel.

Solutions to these issues are discussed on the following pages, including intersection configurations for bicycle lanes, bike boxes, advance stop lines, bicycle-activated detector loops, bicycle specific traffic control signals.

Typical Intersection

Configuration for Bike Lanes

See the Manual on Uniform Traffic Control Devices (MUTCD) for guidance on lane delineation, intersection treatments, and general application of pavement wording and symbols for on-road bicycle facilities and off-road paths (updated version to be released in 2009/2010); example from the MUTCD at right.



Bike Box / Advance Stop Line

A bike box is a relatively simple innovation to improve turning movements for bicyclists without requiring cyclists to merge into traffic to reach the turn lane or use crosswalks as a pedestrian. The bike box is formed by pulling the stop line for vehicles back from the intersection, and adding a stop line for bicyclists immediately behind the crosswalk. When a traffic signal is red, bicyclists can move into this "box" ahead of the cars to make themselves more visible, or to move into a more comfortable position to make a turn. Bike boxes have been used in Cambridge, MA; Eugene, OR; and European cities.

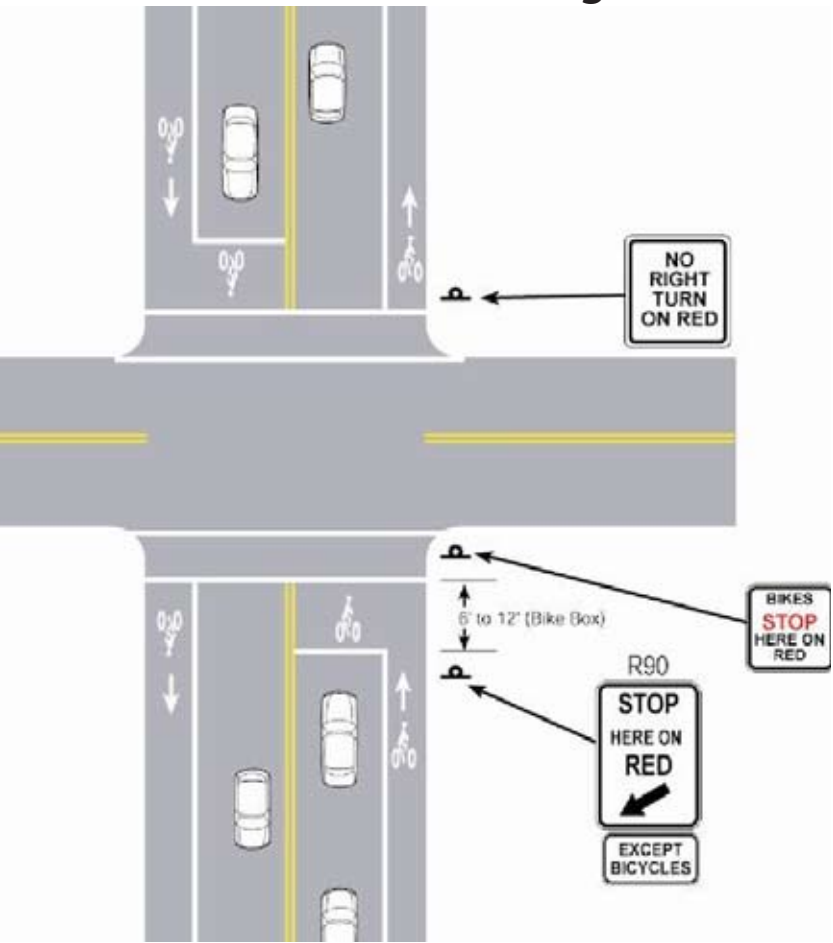
Potential Applications:

- At intersections with a high volume of bicycles and motor vehicles
- Where there are frequent turning conflict and/or intersections with a high percentage of turning movements by both bicyclists and motorists
- At intersections with no right turn on red (RTOR)
- At intersections with high bicycle crash rates
- On roads with bicycle lanes
- Can be combined with a bicycle signal (optional)

Considerations:

- Bike boxes are not currently included in the MUTCD but there are provisions for jurisdictions to request permission to experiment with innovative treatments (and thus with successful application, future inclusion of bike boxes in the MUTCD could occur).
- If a signal turns green as a cyclist is approaching an intersection, they should not use the bike box.
- Motorists will need to be educated to not encroach into the bike box.

Design Guidelines



Plan view of a bike box.



Above and below: Bike boxes filled in with color to emphasize allocation of space to bicycle traffic.



Bicycle Facilities at Railroad Crossings

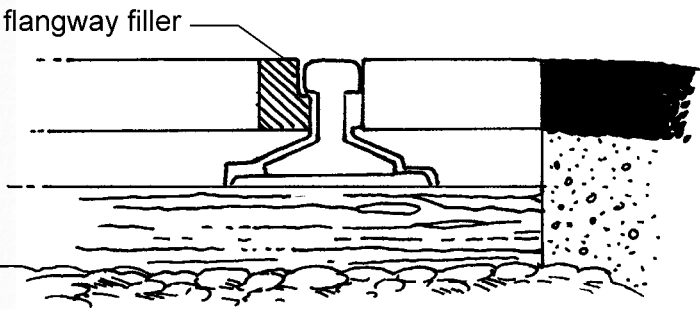
Railroad crossings are particularly hazardous to those who rely on wheeled devices for mobility (railroad crossings have flangeway gaps that allow passage of the wheels of the train, but also have the potential to catch wheelchair casters and bicycle tires). In addition, rails or ties that are not embedded in the travel surface create a tripping hazard. Recommendations:

- Make the Crossing Level: Raise approaches to the tracks and the area between the tracks to the level of the top of the rail.
- Bikes Should Cross RR at Right Angle
- When bikeways or roadways cross railroad tracks at grade, the roadway should ideally be at a right angle to the rails. When the angle of the roadway to the rails is increasingly severe, the approach recommended by Caltrans (Highway Design Manual, Section 1003.6) and AASHTO (Guide for the Development of Bicycle Facilities, 1999, p.60) is to widen the approach roadway shoulder or bicycle facility, allowing bicycles to cross the tracks at a right angle without veering into the path of passing motor vehicle traffic.

- Use Multiple Forms of Warning: Provide railroad crossing information in multiple formats, including signs, flashing lights, and audible sounds.
- Clear Debris Regularly: Perform regular maintenance to clear debris from shoulder areas at railroad crossings.
- Fill Flangeway with Rubberized Material or Concrete Slab: Normal use of rail facilities causes buckling of paved-and-timbered rail crossings. Pavement buckling can be reduced or eliminated by filling the flangeway with rubberized material, concrete slab, or other treatments. A beneficial effect of this is a decrease in long-term maintenance costs.



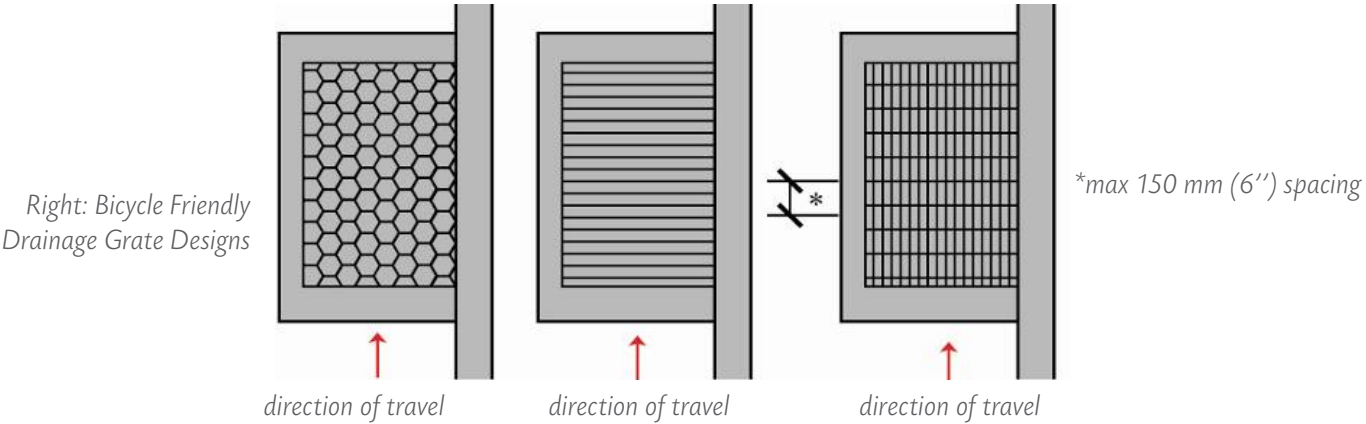
Installing a rubber surface rather than asphalt around railroad flangeways reduces changes in level and other maintenance problems.



The “flangeway filler” eliminates the gap in the path of travel for pedestrians crossing railroad tracks. The filler, consisting of a rubber insert, will deflect downward with the weight of a train and does not affect railway function.

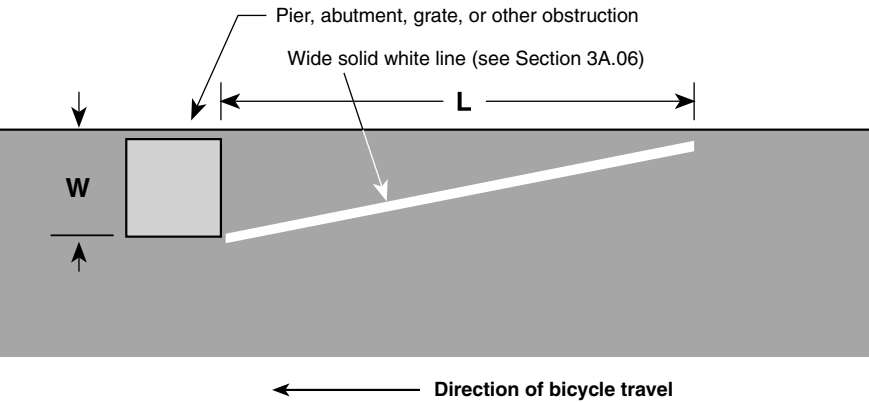
Bicycle Friendly Drainage Grates

Drainage grates usually occupy portions of roadways, such as bicycle lanes, where bicycles frequently travel. Often drainage grates are poorly maintained or are of a design that can damage a bicycle wheel or in severe circumstances, cause a bicyclist to crash. Improper drainage grates create an unfriendly obstacle a cyclist must navigate around, often forcing entrance into a motor vehicle lane in severe cases. Bicycle friendly drainage grates should be installed in all new roadway projects and problem grates should be identified and replaced.



Right: Bicycle Friendly Drainage Grate Designs

Right: MUTCD example of obstruction pavement marking; if dangerous drainage grates (or other obstructions) are not to be fixed in the short term, then this pavement marking should direct cyclists away from the obstruction.



Dangerous Drainage Grate Condition; this example is dangerous due to the grate running parallel to the roadway, creating a trap for bicycle tires.



Dangerous Drainage Grate Condition; this example is dangerous due to the surrounding paving condition (when the road was resurfaced the drainage grate remained at the same height).



Bicycle-Friendly Drainage Grate

Bicycle Access on Transit

Integrating bicycle facilities with transit modes allows bicyclists to greatly expand the area accessible. Below are examples of commuter trains and bus services with customized facilities allowing for simple and secure storage of bicycles without hindering or impeding other passengers. The City of Allentown should continue to accommodate bikes on all buses, and should support similar options if and when light-rail or similar transit options become available.



- 1. Have your bike ready to load—always approach the bus from the curbside. Remove water bottles or other loose items.
- 2. Make eye contact with the driver to alert him/her to your presence.
- 3.If the rack is empty, lift the metal handle and pull the folded bike rack down flat.
- 4. Load the bike in the space nearest the bus.
If another bike is on the rack, load your bike in the open position. You are responsible for loading and securing your bike on the rack. Drivers are not allowed to load or unload bicycles.
- 5. Lift the support arm and hook it over the front tire.
Make sure the support arm clamps the tire and not the fender or frame. Your bike now is securely fastened in the rack.
- 6. Hop on and pay your fare.
- 7. When you reach your stop, tell the driver before you exit the bus that you'll be removing your bike.
Raise the support arm, lower it into place and lift your bike off the rack.
Fold up the rack if it is empty, and step onto the sidewalk with your bike.
NEVER cross in front of the bus—wait until the bus has left the stop.
If the rack is full, please wait for the next bus.

Instructions on how to load a bicycle onto a bus equipped with a bicycle rack, developed for a bicycle user map by Fremont, CA

Bike/Ped Treatments for Transit Stops

Integrating bicycle and pedestrian facilities with transit modes allows users to greatly expand their range of travel or “trip chain”.

- At a minimum, marked crosswalks (especially at mid-block stops), curb ramps, and proper side-walk widths should be provided.
- Although the current buses that serve Allentown are equipped with bicycle racks, bus stops should also incorporate bicycle racks, and at major stops, bicycle lockers.
- Local walking and biking maps should also be provided at bus stops, so that people are aware of the nearby destinations and how best to get there without an automobile.
- Additional elements to consider include: water fountains, pedestrian-scale lighting, legible and adequate transit stop signage, shelter, seating, air compressors, and electronic signs displaying real-time bus arrival information.
- At bus stops, special attention should be paid to the number of lanes and direction of traffic when deciding to locate a stop on the near or far side of an intersection. Also special consideration must be paid to the wheelchair lifts in terms of how and where the mobility impaired will exit and enter the bus.



This typical transit stop features a shelter, ample seating, bicycle parking, landscaping, and trash bins (Image from <http://www.walkinginfo.org>).

Design Guidelines

Sidewalks and Walkways
Sidewalks and walkways are extremely important public right-of-way components often times adjacent to, but separate from automobile traffic. In many ways, they act as the seam between private residences, stores, businesses, and the street.

There are a number of options for different settings, for both downtown and more rural and/or suburban areas. From a wide promenade to, in the case of a more rural environment, a simple asphalt or crushed stone path next to a secondary road, walkway form and topography can vary greatly. In general, sidewalks are constructed of concrete although there are some successful examples where other materials such as asphalt, crushed stone, or other slip resistant material have been used. The width of the walkways should correspond to the conditions present in any given location (i.e. level of pedestrian traffic, building setbacks, or other important natural or cultural features). FHWA (Federal Highway Administration) and the Institute of Transportation Engineers both suggest five feet as the minimum width for a sidewalk. This is considered ample room for two people to walk abreast or for two pedestrians to pass each other. Often downtown areas, near schools, transit stops, or other areas of high pedestrian activity call for much wider sidewalks.

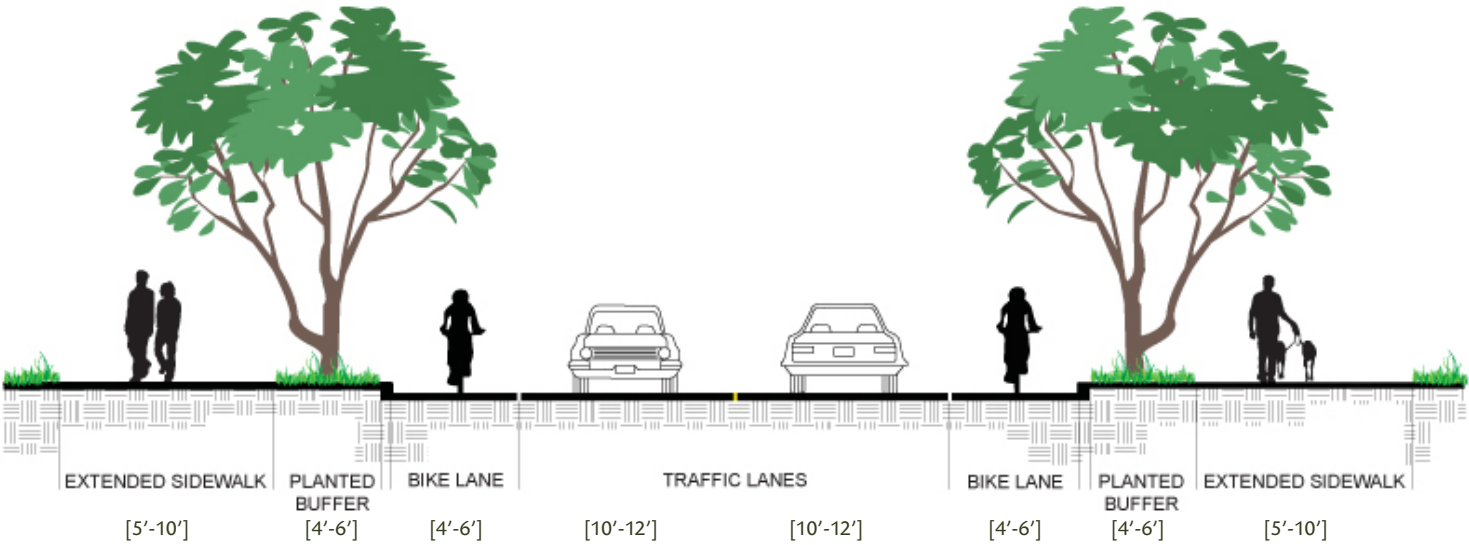


A well designed residential sidewalk will have a width of at least five feet. (Image from <http://www.walkinginfo.org>)

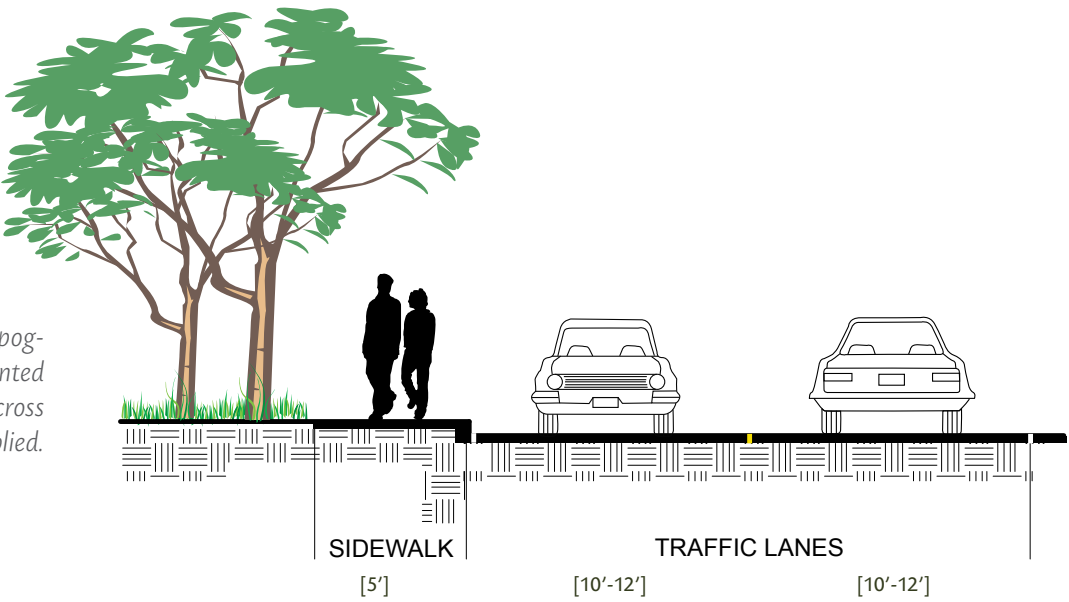


Sidewalk with a vegetated buffer zone. Notice the sense of enclosure created by the large canopy street trees. (Image from <http://www.walkinginfo.org>)

Below: Typical street with bike lanes and adjacent sidewalk.



Right: Where space and topography are limiting and a planted buffer is not possible, this cross section may be applied.



Sidewalks and Walkway Guidelines:

Sidewalk Guideline Sources:

American Association of State Highway and Transportation Officials. (2004). *Guide for the Planning, Design, and Operation of Pedestrian Facilities*.

Metro Regional Government. (2005). *Portland, Oregon: Transportation Information Center*. <http://www.oregon-metro.gov>

Design Guidelines

- Concrete is preferred surface, providing the longest service life and requiring the least maintenance. Permeable pavement such as porous concrete may be considered to improve water quality.
- Sidewalks should be built as flat as possible to accommodate all pedestrians; they should have a running grade of five percent or less; with a two percent maximum cross-slope.
- Concrete sidewalks should be built to minimum depth of four inches; six inches at driveways.
- Sidewalks should be a minimum of five feet wide; sidewalks serving mixed use and commercial areas shall be a minimum of 8 ft in width (12–15 feet is required in front of retail storefronts).
- Buffer zone of two to four feet in local or collector streets; five to six feet in arterial or major streets and up to eight feet in busy streets and downtown to provide space for light poles and other street furniture. See the Vegetation section later in this chapter for shade and buffer opportunities of trees and shrubs.
- Motor vehicle access points should be kept to minimum.
- If a sidewalk with buffer on both sides is not feasible due to topography and right-of-way constraints, then a sidewalk on one side is better than no facility. Each site should be examined in detail to determine placement options.

Curb Ramps

Curb ramps are critical features that provide access between the sidewalk and roadway for wheelchair users, people using walkers, crutches, or hand-carts, people pushing bicycles or strollers, and pedestrians with mobility or other physical impairments. In accordance with the 1973 Federal Rehabilitation Act and to comply with the 1990 Federal ADA requirements, curb ramps must be installed at all intersections and mid-block locations where pedestrian crossings exist (Pedestrian and Bicycle Information Center: <http://www.walkinginfo.org/engineering/roadway-ramps.cfm>). In addition, these federal regulations require that all new constructed or altered roadways include curb ramps.

Two separate curb ramps should be provided at each intersection (see image below). With only one large curb ramp serving the entire corner, there is not safe connectivity for the pedestrian. Dangerous conditions exist when the single, large curb ramp inadvertently directs a pedestrian into the center of the intersection, or in front of an unsuspecting, turning vehicle.

Curb Ramp Guidelines:

- Two separate curb ramps, one for each crosswalk, should be provided at corner of an intersection.
- Curb ramp should have a slope no greater than 1:12 (8.33%). Side flares should not exceed 1:10 (10%); it is recommended that much less steep slopes be used whenever possible.

Curb Ramp Guideline Sources:

Metro Regional Government. (2005). Portland, Oregon: Transportation Information Center. <http://www.oregon-metro.gov>

Left: The curb ramps shown have two separate ramps at the intersection (visible across the street) (Image from <http://www.walkinginfo.org>).

For additional information on curb ramps see Accessible Rights-of-Way: A Design Guide, by the U.S. Access Board and the Federal Highway Administration, and Designing Sidewalks and Trails for Access, Parts I and II, by the Federal Highway Administration. Visit: www.access-board.gov for the Access board's right-of-way report.



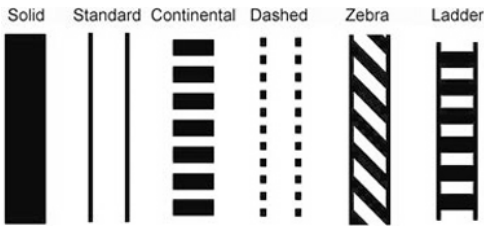
Marked Crosswalks

A marked crosswalk designates a pedestrian right-of-way across a street. It is often installed at controlled intersections or at key locations along the street (a.k.a. mid-block crossings). Every attempt should be made to install crossings at the specific point at which pedestrians are most likely to cross: a well-designed traffic calming location is not effective if pedestrians are instead using more seemingly convenient and potentially dangerous location to cross the street. Marked pedestrian crosswalks may be used under the following conditions: 1) At locations with stop signs or traffic signals, 2) At non-signalized street crossing locations in designated school zones, and 3) At non-signalized locations where engineering judgment dictates that the use of specifically designated crosswalks are desirable.

There is a variety of form, pattern, and materials to choose from when creating a marked crosswalk. It is important however to provide crosswalks that are not slippery, are free of tripping hazards, or are otherwise difficult to maneuver by any person including those with physical mobility or vision impairments. Although attractive materials such as inlaid stone or certain types of brick may provide character and aesthetic value, the crosswalk can become slippery. Potential materials can be vetted by requesting case studies from suppliers regarding where the materials have been successfully applied. Also, as some materials degrade from use or if they are improperly installed, they may become a hazard for the mobility or vision impaired.

Crosswalk Guidelines:

- Should not be installed in an uncontrolled environment [at intersections without traffic signals] where speeds exceed 40 mph. (AASHTO, 2004)
- Crosswalks alone may not be enough and should be used in conjunction with other measures to improve pedestrian crossing safety, particularly on roads with average daily traffic (ADT) above 10,000
- Width of marked crosswalk should be at least six feet; ideally ten feet or wider in downtown areas.
- Curb ramps and other sloped areas should be fully contained within the markings.
- Crosswalk markings should extend the full length of the crossings.
- Crosswalk markings should be white per MUTCD.
- Either the 'continental' or 'ladder' patterns are recommended for intersection improvements for aesthetic and visibility purposes. Lines should be one to two feet wide and spaced one to five feet apart.



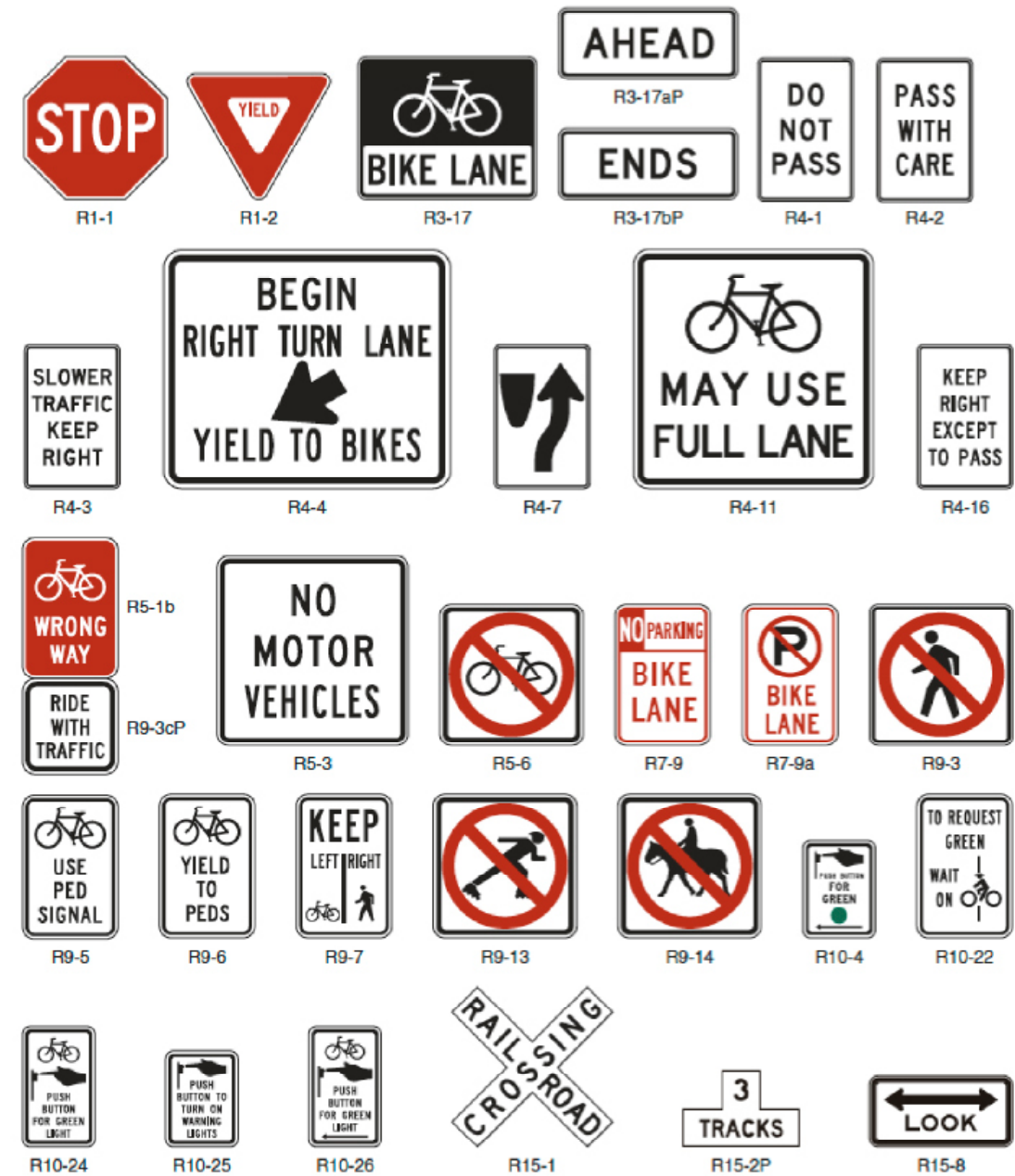
A variety of patterns are possible in designing a crosswalk; an example of a 'continental' design is shown above.

Crosswalk Guideline Sources:

American Association of State Highway and Transportation Officials. (2004). Guide for the Planning, Design, and Operation of Pedestrian Facilities.

Metro Regional Government. (2005). Portland, Oregon: Transportation Information Center. <http://www.oregon-metro.gov>

Figure 9B-2. Regulatory Signs and Plaques for Bicycle Facilities



On-Road Regulatory Signs for Bicycle Facilities - Manual of Uniform Traffic Control Devices for Streets and Highways (MUTCD) 2009 Edition

The Federal Highway Administration's MUTCD has a palette of regulatory signs for utilization of bicycle and pedestrian circulation with motor vehicles. In general, these devices should be used in conjunction with approved striping techniques. As a general rule some of these signs, such as R9-3, should not be used if possible because they do not present a pedestrian and bicycle friendly image to the general public.

Potential Off-Road Trail Signing Icons:

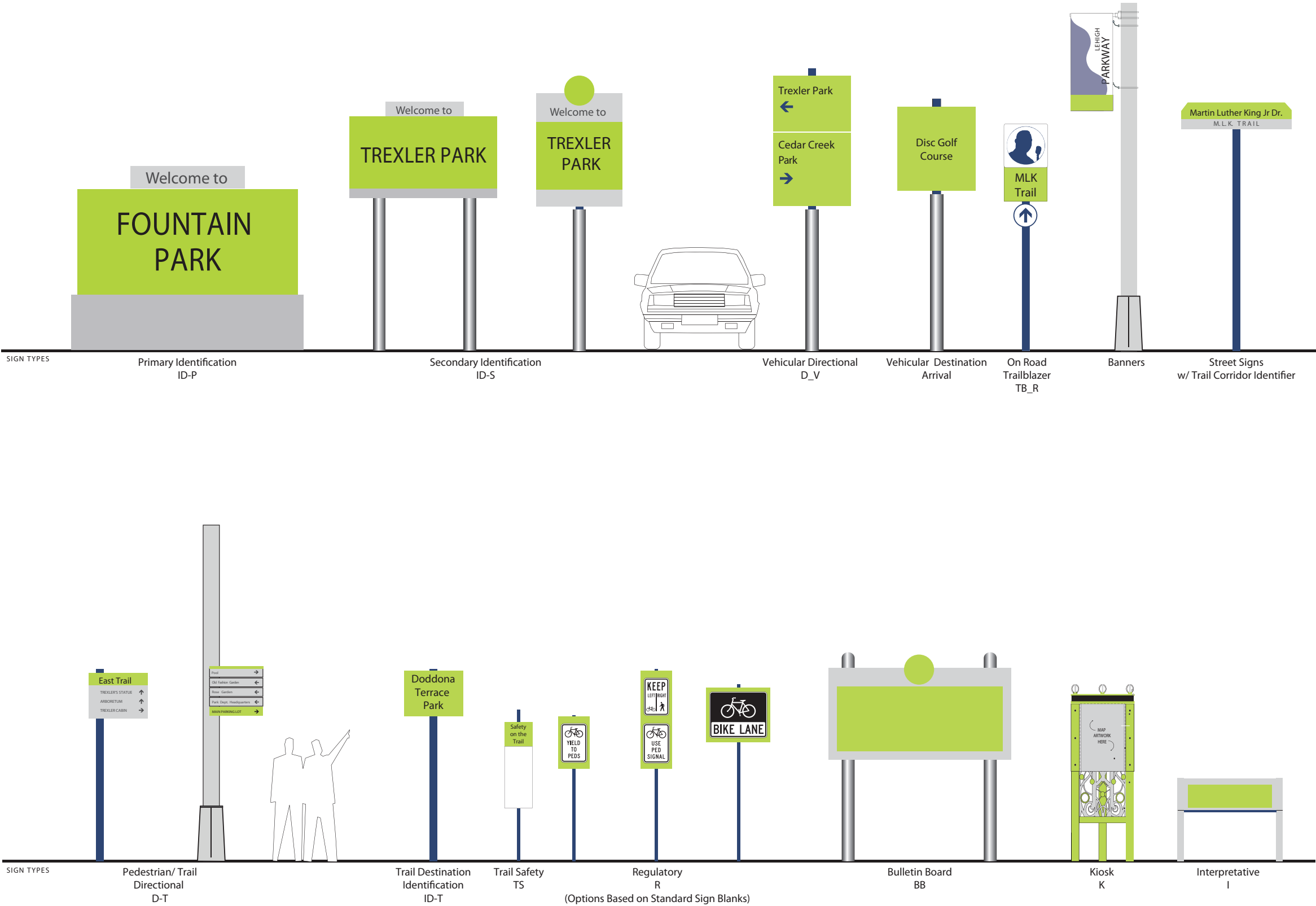
Off-road trail signing allows for greater design flexibility. One option to reduce the size of the signs and to minimize text, is to utilize icons as a way to provide basic information. Any icons that are utilized should be simple and their intent should be very obvious and universal. The above is an example palette that covers typical messages for trail users.

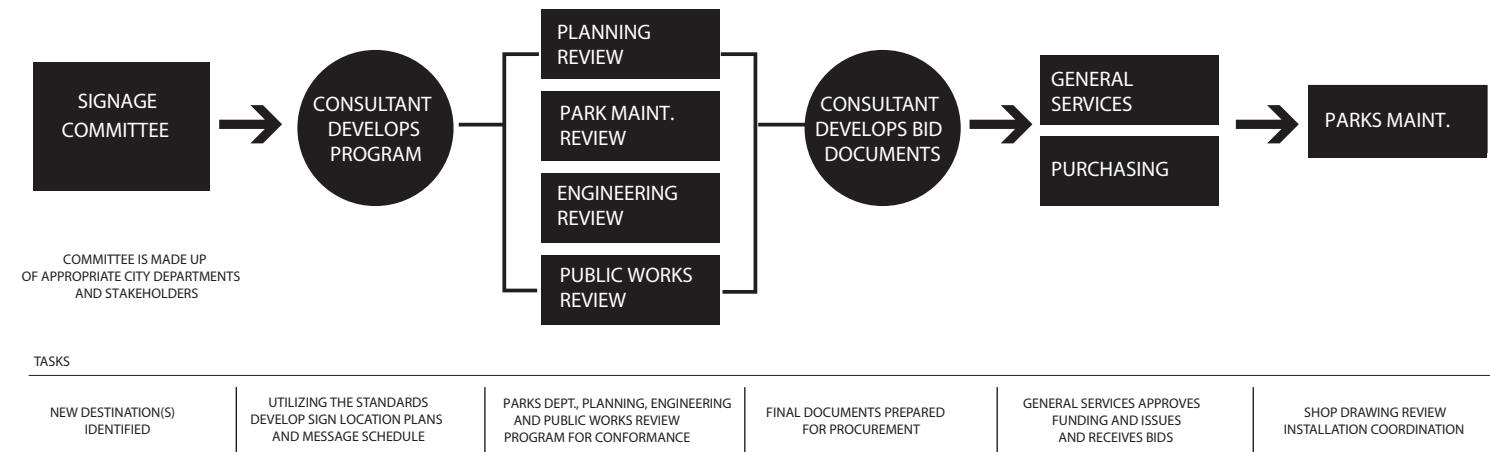
The Value of a Comprehensive Signing System

One of the primary results of a coordinated sign program is that it naturally reduces clutter by presenting a consistent design and organized information. The sign program will reduce the need for multiple signs at congested decision points. One sign holds multiple messages, rather than three signs arbitrarily mounted to a pole(s). Consistent design: Standard colors, graphics, typefaces and size help to present a well-planned park system and trail network and a uniform identity for the City as a whole. Sign placement is planned as part of a comprehensive system, taking into consideration both the environment and type of circulation and targeted audience. Placement is also logical and based on a coordinated trail or park-wide sign system. Signs can be sized depending on their context and pedestrian and trailblazer sign types can be mounted to existing poles whenever possible. This will reduce the quantity of poles added to current conditions as well as reduce obstacles for cyclist and pedestrians. With regards to regulatory signs, standard sign blanks can be used for the panels in order to keep costs manageable and ease replacement, however, they should be mounted onto a consistent system backing panel that are painted the same color on both front and back, as well as the pole and mounted bracket. This will provide a high-level of consistency even though regulatory signs will vary dramatically in terms of size and color, etc.

Additional Recommendations:

Purge and Repair: All unnecessary, damaged or inappropriate park/street furniture should be removed or replaced. If not already established a “Design Standard Manual” should be set (and utilized) for all park/street furniture. This would provide a variety of coordinated options throughout the park system based on individual park aesthetics and environments. Elements may include: lamp posts, bike racks, benches, trash cans, tree grates, etc.





Maintenance Matrix for Permanent Signs

Sign Longevity	0-4 Years	5-9 years	9-25 years
Design and Planning	Extensive design and planning pro-gram continues even after sign system in place. Client plays a crucial role	Moderate amount of design and planning. Less input needed from the client.	One time design and planning costs.
Sign System	Light attachment details. Flexible System. Extensive computerized sys-tem schedule.	Attachment details allow for some replacement. Com-puter database for sign changes	Durable attachment. Very difficult to re-move. No database needed for system.
Materials	Low grade materials	Medium Grade ma-terials.	High quality materi-als
Changeability	Limited Changeabil-ity. Signs need little demountability of parts but signs must be easy to remove.	Moderate Change-ability. Extensive demountability of sign parts especially the sign face.	Complete Change-ability. Every part of the sign must be fitted with removable parts for changes in the system
Cleaning	No major investment in cleaning	Major cleaning schedule	Major cleaning schedule
Replacement	Replacement sched-ule on a month-by-month basis.	Yearly replacement schedule	Bi-yearly replace-ment schedule
Management	Extensive daily inter-play between client and fabricator	Monthly interplay between client and fabricator	Monthly interplay between client and fabricator

This information was developed with assistance from Craig Berger of the Society for Environmental Graphic Design and MERJE Environments and Experiences.

The Sign

Foundations: Choose the foundation system that fits the environment. Embedded for soil and park conditions. Slip Base or Sleeve for high traffic areas. Remember to have a clear foundation removal and replacement plan in place before selecting a foundation system.

Pole: Pole color and paint should be selected based on location. Use of existing poles is possible in very select conditions and only with clear structural engineering drawings. Poles get the most graffiti damage so have a continuous cleaning and repainting program in place.

Panel: Should have a higher-level professional cleaning program in place. Panels in high sun or tougher areas should be coated with a film, (Tedlar, mylar) to protect against ultraviolet rays. Cleaning should be carefully done with a lemon soap for stickers, and a mild paint thinner for graffiti. Professionals or highly trained staff should complete panel cleaning.

Special Considerations

Windloads:The signs should be engineered to withstand Monsoon force winds of a minimum of 90 mph.

Availability of Materials: All sign components are made of industry standard materials and fabrication techniques. These include: 1/4 thick aluminum sheet, 4” tube poles, reflective vinyl, automotive grade paint.

In-House Fabrication: Will likely be limited to graphics and lettering for changing messages, poles, foundations, and installation. Initial “attic stock” of parts should be included in the base bid of each phase of the project. Attic stock can include poles (painted), sign panels (painted/no lettering), brackets finished and painted and other parts .

Historic Compatibility: Signs are designed to coordinate with the historic nature of the area. Design considerations include size, scale, color, placement, and quantity of signs. Future signage additions should adhere to the design philosophy established by the sign program.

Best Cleaning Practices

Maintenance Schedule: Signs should be cleaned at lest annually, twice a year is preferred.

Dirt and Grime: A mix of Simple Green and water

Removing Graffiti: Mild Enamel Thinner

Removing Stickers: Goof Off / Goo Gone

Lettering: For gateways or panels that will not be changed often a complete fi lm coating or silk-screening works best. For signs that will change a durable vinyl lettering with a clear program for changing information. Panels and letter placement should be designed to permit ease of changing information (Standard message heights, etc.)